In the name of God

Bone Cement

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Rasaneh Takhassosi Publishing - 1404

سرشناسه : حسن‌زاده چایجانی، ذبیح‌الله، ۱۳۴۳
Hassanzadeh Chayjani, Zabihollah, 1964
عنوان قراردادی : سیمان استخوان .انگلیسی
عنوان و نام پدیدآور : .Bone cement[Book]/ author Zabihollah Hassanzadeh Chayjan
مشخصات نشر : تهران : رسانه تخصصی ، ۱۴۰۴= ۲۰۲۵م.
مشخصات ظاهری : ۱۳۶ص.
شابک : ۳-۷۳۵-۲۹۳-۶۰۰-۹۷۸
وضعیت فهرست نویسی : فیپا
یادداشت : زبان: انگلیسی.
موضوع : جراحی ارتوپدی
موضوع : Orthopedic surgery
موضوع : جراحی ارتوپدی -- راهنمای آموزشی
موضوع : Orthopedic surgery -- Study and teaching
موضوع : سیمان استخوانی
موضوع : Bone cements
موضوع : مواد چسبنده در جراحی
موضوع : Adhesives in surgery
رده بندی کنگره : RD۷۳۱
رده بندی دیویی : ۶۱۷/۳
شماره کتابشناسی ملی : ۱۰۲۴۷۶۳۶

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Author:
Dr. Zabihollah Hassanzadeh Chayjani
Publisher: **Rasane Takhassosi**
Designer and Layout:
Hamed Kazemi – Najmeh al-Sadat Mirshafiei
Printing turn : **First /2025 (1404)**
ISBN: **978-600-293-735-3**
Circulation: **500**
Price: **450/000 Tomans**

Adress: Unit 1, No. 8
Kandovan alley, Enghelab Street
Ferdowsi Square
RASANE TAKHASSOSI PUBLISHING
Tel: +982166737133-66737332
+989123049109

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Preface

1. Bone cement has held a special place in both modern and classical orthopedics for many years. Its applications and related techniques have addressed many orthopedic challenges and significantly improved surgical outcomes.
2. This field has not only made remarkable progress in scientific production—with the publication of thousands of articles and specialized theories—but has also claimed a considerable share in the trade and medical equipment industry (Archir Klin Chi 18,1).
3. Different generations of bone cement have been designed and produced with diverse compositions and characteristics so that this widely used biomaterial remains both safe and effective.
4. According to numerous studies, the most common applications of orthopedic cement are in arthroplasty (joint replacement), the treatment of bone infections, and also in spinal procedures, particularly in the treatment of kyphosis and vertebroplasty.
5. However, the potential complications and challenges associated with the use of bone cement during surgery and throughout the treatment process are also serious issues that will be discussed in detail later in this book.
6. The importance of familiarizing the new generation of orthopedic specialists with this subject—especially considering the omission or marginalization of bone cement topics in many academic resources—provided a valuable opportunity to compile and edit this work.

Acknowledgment

I feel obliged to express my gratitude for the valuable efforts of Ms. Najmeh-al-Sadat Mirshafiei in the scientific and content editing of this work. I also sincerely thank Mr. Ashtiani, the successful director of the specialized media publishing house, whose constant pursuit has been to provide the best for the scientific and medical community.

Respectfully,

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Chapter 1

Biomaterials

Chapter 1: Biomaterials

Definition of Biomaterial

Biomaterials are substances, either natural or synthetic, that are designed and engineered to interact with biological systems in such a way that they can perform therapeutic, reparative, or substitutive functions for body tissues.This field is an interdisciplinary science that integrates biology, medicine, metallurgy, materials engineering, nanotechnology, 3D printing technology, and pharmacology.

Biomaterials have a history of over 2,000 years in medical applications, ranging from primitive materials such as ivory and animal bone to today's advanced biomaterials, which have consistently played a role in patient treatment. At present, these materials are widely used across various medical disciplines, including dentistry, orthopedics, cardiovascular surgery, ophthalmology, and reconstructive and plastic surgery.

Types of Biomaterials

Biomaterials are classified into two main categories:

1. Natural Materials: such as ivory, gold, pine resin, and other substances derived from nature.
2. Synthetic Materials: such as polymers, ceramics, and metal alloys.



Fig 1. Natural materials



Fig 2. Synthetic materials

Medical Applications

Biomaterials are employed both inside and outside the body. Some of their key applications include:

- **Dentistry:** The use of gold for dental restorations in ancient civilizations such as China, Rome, and among the indigenous peoples of Mexico and Iran.
- **Orthopedics:** The use of ivory and pine resin to construct the earliest wooden total knee arthroplasty (TKA) prostheses.
- **Artificial Prostheses:** Fabrication of artificial joints, orthotic devices, mechanical heart valves, intraocular lenses, and spinal instrumentation.

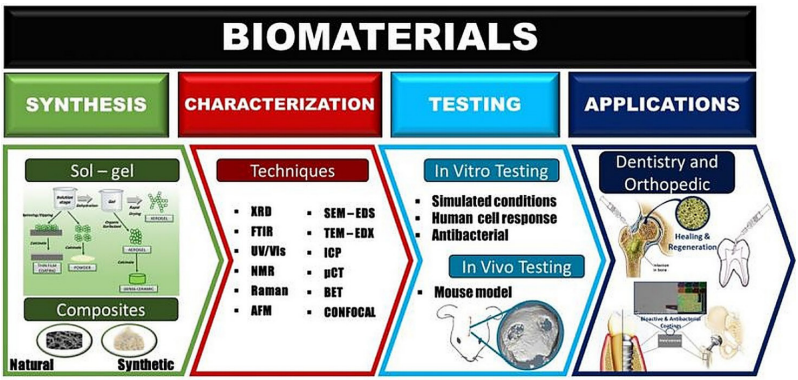


Fig 3. Medical Applications of Biomaterials

Desirable In Vivo Biomaterials

Desirable biomaterials should possess the following characteristics:

- **Biocompatibility:** Elicits minimal immune or inflammatory response.
- **Adequate Mechanical Strength:** Withstand physiological loads without failure.
- **Tissue Integration:** Promote stable bonding or ingrowth with surrounding tissues.

Today, a significant portion of healthcare spending is dedicated to the research, development, and deployment of advanced biomaterials.

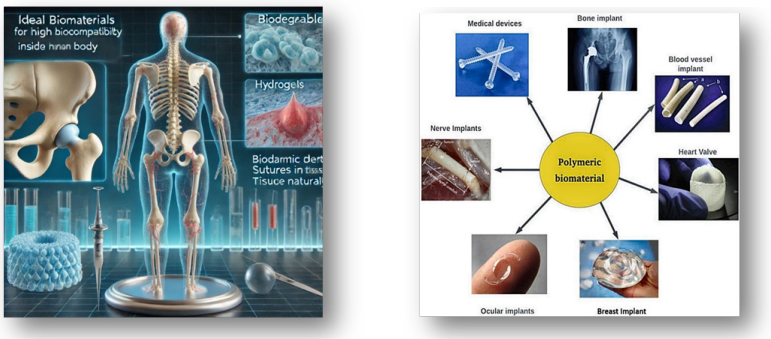
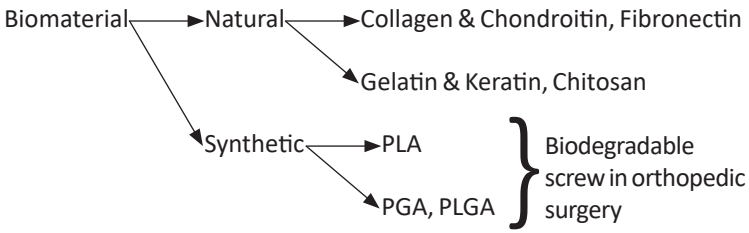


Fig 4. Applications of Biomaterials

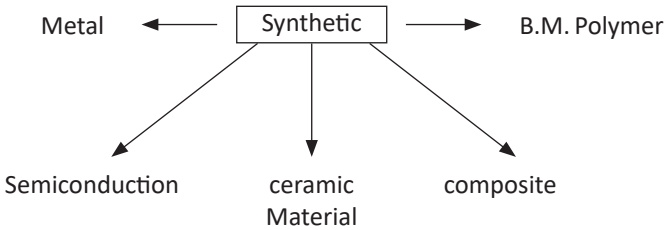
Applications of Biomaterials

Category of Biomaterials	Example Materials	Medical Applications
Metals	Titanium, Stainless Steel, Cobalt-Chromium Alloys	Orthopedic implants, dental implants, and artificial heart valves
Polymers	Polylactic Acid (PLA), Silicone, Polymethyl Methacrylate (PMMA)	Biodegradable sutures, contact lenses, and artificial heart components
Ceramics	Hydroxyapatite, Zirconia, Alumina	Bone substitution, dental implants, and artificial joints
Composites	Carbon fiber-reinforced materials, ceramic-reinforced polymers	Prosthetic limbs, reinforcement of orthopedic implants
Hydrogels	Polyethylene glycol-based hydrogels	Tissue engineering, wound healing, and drug delivery

Types of Biomaterials:



Types of Synthetic Biomaterials:



Synthetic Biomaterials and Their Types

Synthetic (man-made) biomaterials are generally classified into four main categories:

1.Metals:

- Used in spin and orthopedic implants.
- Gold is utilized in dental applications.

2.Semiconductors:

- Applied in microelectrode implants.
- Used in biosensors.

3.Polymers:

- Orthopedic bone cements.
- Ocular (eye) implants.
- Artificial skin and cartilage.

4.Ceramics:

- Hip prostheses.

- Artificial heart valves.

Biomaterials Science

Biomaterials require advanced knowledge in various domains and rely heavily on interdisciplinary research. This field is closely linked with disciplines such as biology, materials chemistry, nanotechnology, drug delivery systems, tissue engineering, and metallurgy.

Specific Applications of Biomaterials in Modern Medicine

Biomaterials are used in a wide range of applications in contemporary medical practice, including:

- Vascular grafts
- Coatings on medical devices
- Catheters
- Implants (especially drug delivery systems)
- Cranioplasty
- Artificial heart
- Dialysis tubes
- Vertebroplasty
- Treatment of osteomyelitis
- Joint arthroplasty
- Management of bone tumors
- Membranes for oxygenation and detoxification
- Injectable drug delivery and imaging systems
- Membranes and porous scaffolds for tissue regeneration applications



Fig 5. Special Medical Applications of Biomaterials

Q&A – Questions and Answers

1. Where are biomaterials used?

Medical-grade biomaterials are used both inside and outside the body. They are employed in various applications such as cardiac stents, prostheses, hearing aids, and orthoses used in limb amputations.

2. Why do we need biomaterials?

Due to the deterioration or dysfunction of body organs or the need for assistance in therapeutic procedures (e.g., intravenous catheters), biomaterials are widely applied in diagnostic, therapeutic, and healthcare-related fields.

3. What general characteristics must biomaterials have to be suitable for medical use?

1.Performance: Ability to fulfill the intended function.

2.Process: Appropriate synthesis and manufacturing methods.

3.Physical and biological properties: Features such as mechanical strength and compatibility with body tissues.

4. What characteristics are required for biomaterials used inside the body?

1.Biocompatibility: Ability to coexist with body tissues without eliciting adverse reactions.

2.Bioinertness: Chemical inertness in blood and serum.

3.Non-toxicity: Absence of short-term or long-term toxic effects.

4.Biomechanical properties: Mechanical compatibility with the targeted anatomical structure.

5. How does the body respond to biomaterials?

Immune cells—including T cells, B cells, and macrophages—react to biomaterials. These responses may include:

- **Fibrosis:** Formation of fibrotic tissue around the biomaterial.
- **Protein attachment:** Adsorption of host proteins onto the biomaterial surface.
- **Rejection:** Adverse immune response that may lead to biomaterial failure.
- **Coagulopathy:** Disruption of normal blood coagulation mechanisms.
- **Loosening, erosion, degradation:** Mechanical failure or material breakdown within the body.

6. What is the concept of biocompatibility?

Williams (1987) defined biocompatibility as the ability of a material to perform with an appropriate host response in a specific application.

> "Biocompatibility is the ability of a material to perform with an appropriate host response in a specific application."

In other words, a biocompatible material does not induce inflammation, allergic reactions, immune responses, or carcinogenic effects. The required properties of biomaterials vary depending on their intended application; for instance, anticoagulant properties may be essential in vascular tissues, whereas molding capability is more critical in cosmetic applications.

Summary of Biomaterial & Body Reactions

Interaction	Result	Material
Toxic	Tissue death	Most of the metals
Nearly Interaction	Fibrous tissue formation	Ceramics like Zirconium
Bioactive	Surface tissue	Hydroxyapatite bio glass
Degradable	Substitution of material with tissue	Tricalcium phosphate Collagen

Comprehensive References for Further Study:

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Kim, J. (2023). Investigating immune modularity in biomaterials. Advanced Healthcare Materials, 12(4), 2201234.

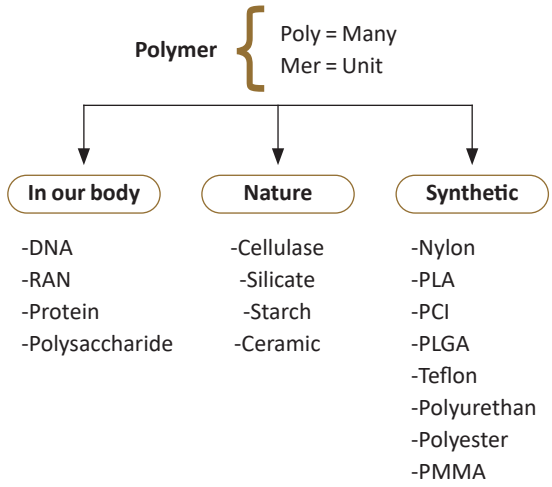
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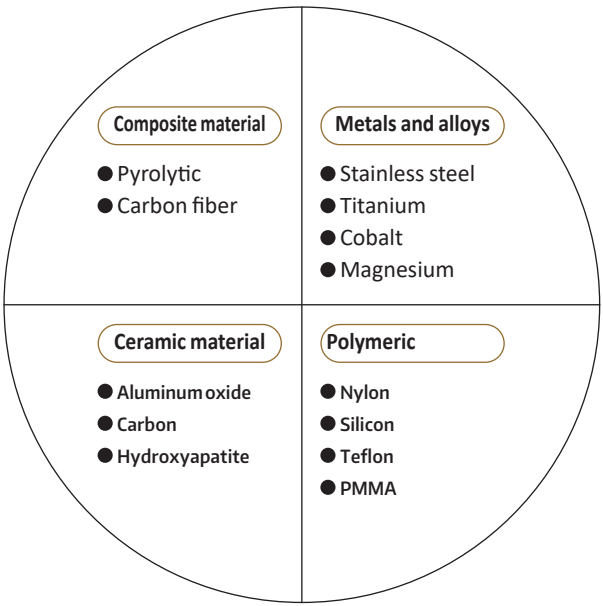
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Chapter 2

Polymers

Chapter 2: Polymers





Chapter 3

History of Bone Cement

Chapter 3: History of Bone Cement

Historical Evolution of Medical Treatment and Management of Advanced Osteoarthritis

Early Attempts at Total Knee Arthroplasty (TKA).

The first attempt at knee joint replacement was made in 1870 by a German surgeon named Themistokles Gluck. He designed and implanted a knee prosthesis using ivory, colophony (a natural resin), and plaster in a patient. Although this effort ultimately failed, it made a significant contribution to the historical development of total hip arthroplasty (THA).

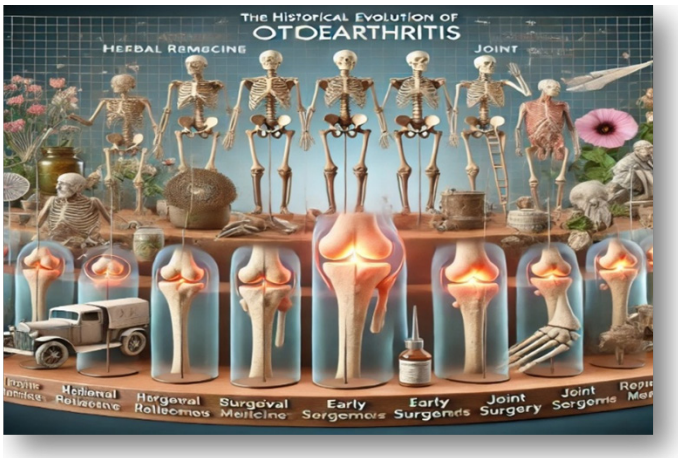


Fig 6. Historical Evolution of Medicine and Treatment of Advanced Osteoarthritis

The Birth of Orthopedic Cement

- **1901: Synthesis of Polymethyl Methacrylate (PMMA)**

German chemist Otto Rohm successfully synthesized polymethyl methacrylate (PMMA).

- **1936: Development of a New PMMA Formulation**

Kulzer Co. introduced a new heat-curing PMMA formulation that polymerized at 100°C, marking a significant advancement. However, the exothermic polymerization reaction posed a major challenge.

- **1943: Application of PMMA in Cranial Surgeries**

By modifying the composition of PMMA, it was first used in surgeries to repair skull defects. This success marked the beginning of PMMA applications in dentistry and cranioplasty.

- **1940s: Expansion of Dental Applications**

Breusch Kuhn extended the use of PMMA in dental treatments.

- **1940–1953: Orthopedic Applications of PMMA**

Judet was the first to propose the use of PMMA in hemiarthroplasty.

- **1953–1958: First Use of PMMA in Total Hip Arthroplasty (THA)**

Sir John Charnley, an English surgeon, was the pioneer in using PMMA for hip joint replacement in Manchester.

Dental ⟶ Spinal-Orthopedic ⟶ Oncology ⟶ Infection & New Versions in Arthroplasty

Wide Applications of Orthopedic Cement

Orthopedic cement has been utilized across various fields, including dentistry, spinal and orthopedic surgeries, oncology, infection treatment, and newer arthroplasty techniques.

Major Advances in the Development of Orthopedic Cement

- Addition of radiopaque materials to enable visualization of the cement on radiographic images.
- Improvement of curing speed and cement stability.
- Incorporation of antibiotics to reduce the risk of infection.
- Enhancement of the modulus of elasticity (ME) to increase mechanical strength.
- Quality improvements by major manufacturers (e.g., green pigment for chlorophyll-based coloring in revision surgeries to differentiate bone from cement).

Key Scientists in the Development of Orthopedic Cement

1. Professor Pechmann: At the University of Tübingen, Germany, he taught polymerization to his student Otto Rohm.

2. 1966: CMW company produced the first official orthopedic cement product.

3. 1996: Cement sterilization using gamma radiation was implemented.

4. 2001–2003: The U.S. Food and Drug Administration (FDA) approved bone cement for medical applications (510(k) clearance).

5. 2006: The second and third generations of orthopedic cement were introduced.

Generations of Bone Cement (PMMA)

Bone cement (PMMA), like many other products, has undergone an evolutionary and progressive development, with quality improvements over time. Major cement manufacturers, through continuous innovation, have maintained their market positions against competitors. These innovations include formulation improvements, elimination of weaknesses, and gaining superior production capabilities. This progress reflects a successful integration of industry and medical science.

Complications of Orthopedic Cement:

- **Can allergy to PMMA or its components be diagnosed or predicted?**

Can allergy to polymethyl methacrylate (PMMA) or its components cause problems for the patient? Does this sensitivity contribute to early loosening?

- **Is orthopedic cement potentially carcinogenic, or does it increase the risk of malignancy?**

Clinical Reports and Studies:

1. **Haddad and Cob (1996) Report:**
   These researchers reported 7 patients who experienced rapid loosening after cemented total hip arthroplasty (THA). Investigations revealed that these patients had severe hypersensitivity to the liquid component of the cement and N, N-dimethyl para-toluidine, a polymerization catalyst.
2. **Delayed Hypersensitivity Reaction Model (Guin, 1995):**

Individuals with genetic predispositions, such as contact dermatitis, may develop hypersensitivity reactions during the cementing process. These reactions can be triggered by exposure to cement components or during polymerization, occurring via chemical mediators or cellular responses (Dalha, 1979; Goodwin, 1976).

3. **Association Between Dermatitis and Early Loosening:**
   Some studies have shown a link between dermatitis and early prosthesis loosening. Therefore, it is recommended to obtain detailed histories in patients with allergies, eczema, or respiratory reactions and perform skin patch tests with agents like methyl methacrylate (MA-100). The test site (usually on the back) is evaluated 72 hours after exposure (Kevin Kaplan, 2002).
4. **Clementid Study (1980):**
   In a study of 42 patients undergoing cemented hip arthroplasty, 25% showed positive skin patch tests within 6 months postoperatively.
5. **Sources of Allergy:**

Allergy may result from skin contact with cement liquid (especially if gloves are compromised) or inhalation of released monomers. Concentrations of these substances in the blood of patients and medical staff increase immediately after exposure (Buchhron, 1992). These substances can irritate the skin, respiratory tract, eyes, oral mucosa, and gastrointestinal system.

(Lozewicz et al., 1985; Savonius et al., 1993; Marshal et al., 1978; Gift, 1998)

Bone Cement and Carcinogenic Effects:

1. Carcinogenicity tests on animal models have been negative

Genotoxicity Study of Bone Cement in Animal Models

Studies have shown that bone cement does not induce reverse mutations in genes or chromosomes. These findings were published in the Chinese Journal of Medical Instrumentation (2011, Volume 35, Issue 6, pp. 425–427).

2. Systemic and local toxic effects exist.

Systemic and Local Toxicity of MMA

Methyl methacrylate (MMA), a component of bone cement, has systemic and local toxic effects (Kindt, 1995; Lewis, 1997).

3. Cement exhibits both acute and chronic effects in dentistry.

Acute and Chronic Effects of Dental PMMA

PMMA used in dentistry can have both acute and chronic effects.

4. Target organs in allergic reactions include the skin, eyes, and mucous membranes.

Allergic Reactions

- Skin, eye, and mucous membrane irritation.
- Respiratory symptoms such as coughing and wheezing.
- Asthma attacks.

5. Headache and drowsiness may occur.

Neurological Symptoms

- Headaches.
- Neurological symptoms such as lethargy and difficulty concentrating.

6. Cardiopulmonary effects are significant.

Chronic Effects

- Reduced lung function.
- Nasal symptoms.
- Cardiovascular effects.

Comprehensive References for Further Study:

1. Gosovi, S. (2010). Local and systemic effects of PMMA. Dental Research Journal, 7(2), 82-87.

2. Ciapetti, G. (2000). Free radical production by benzoyl peroxide (BPO) in bone cement powder and long-term genotoxicity concerns.

Skin Burns Caused by Bone Cement in Total Hip Arthroplasty (THA)

Case Series from Japan (Murayama K et al)

Contact of the skin with bone cement components during the exothermic polymerization phase can cause burns. Areas such as the heel, malleoli, and lower leg are particularly at risk.

Between September 1 and October 1, 2018, a total of 841 THA procedures using the direct anterior approach (DAA) were performed. Cemented stems were used in 312 cases, while all cups were uncemented. None of the patients had a prior history of skin diseases, allergies, or related complaints.

The mean operative time was 37 minutes for unilateral and 88 minutes for bilateral surgeries. Four cases of burns were reported on the medial or lateral thigh and heel, likely caused by contact with cement fragments or cement infiltration into the canal.

Burston B (2007) reported cement-related burns and noted that increased cement volume and loose cement fragments beneath dressings can cause thermal injuries.

Temperature Increase and Cement Thickness:

- Laboratory studies have recorded temperature rises to 100°C immediately after mixing cement, whereas in vivo temperatures typically range between 38–50°C (Weber E, 1976; Huiskes R, 1980; Toksvig L, 1991; Lic, 2003; Reckling, 1977).
- The highest temperatures occur at the thickest cement layers:
  - 41°C at 1 mm thickness
  - 56°C at 5 mm thickness
- The ideal and usual cement thickness is 2–3 mm.

Risk Factors for Cement-Related Burns:

Although skin burns are not common, they can occur under the following conditions:

- Cement fragments protruding from the canal compressed under elastic bandages, especially if the thickness exceeds 10 mm.
- Loose cement particles in direct contact with the skin under the dressing.

Reference

Journal of Clinical Case Reports. (2019). ISSN: [Insert ISSN Number], 9(2).

Non-Carcinogenicity of Bone Cement in Human Models:

Is PMMA Carcinogenic?

Methyl Methacrylate (MMA) is widely used in the production of glass, adhesives, paints, and in dental and orthopedic procedures (e.g., arthroplasty).

Human Studies

- **No Carcinogenic Effects in Humans:** A study by Tomenson et al. (2005) found no evidence of carcinogenicity in humans.
- **Occupational Exposure Limits:** The European Commission has set indicative occupational exposure limit values (IOELVs) for MMA at a maximum of 50 ppm over 8 hours, with a short-term exposure limit not exceeding 100 ppm over 15 minutes.

Animal Studies

- **Carcinogenic Effects in Animals:** A study by Contze et al. (1967) reported carcinogenic effects in animal models.
- **Case Report:** Thompson and Fntin (1969) documented a case of carcinogenicity following 186 applications of bone cement.

The Use of Bone Cement Was Approved by the FDA in 2003.FDA Approval

The U.S. Food and Drug Administration (FDA) has classified PMMA as safe and non-carcinogenic (Class II controlled substance).

Clinical Case Report

Percussion in a Young Patient: A case report published in the International Journal of Research in Orthopedics (Kohan, 2020, Volume 6, Issue 1, pp. 230–232) discussed a young patient who experienced complications after a long fall.

Chapter 4

Orthopedic Cement and Its Components

Chapter 4:
Orthopedic Cement and Its Components

Orthopedic cement consists of two main components: a liquid (monomer) and a powder (polymer). Physically, the monomer—methyl methacrylate (MMA)—is a colorless, transparent, and flammable substance. The powder contains various additives, including radiopaque agents, pigments, antibiotics, and more.

Comparative Study of Cement Mixing Techniques

A research study compared two methods of mixing orthopedic bone cement:

1.Handmade Mixing

2.Vacuum Mixing Technique



Fig 7. Handmade Mixing



Fig 8. Vacuum Mixing

The results of this study showed that:

- No Statistically Significant Difference: There was no statistically significant difference in the final concentration or composition of the cement produced by the two methods.
- Short-Term Complications: No notable difference in the incidence of short-term complications was observed between the two mixing techniques.

Jasmin, J., et al. (2014). Ann.Cocep, 58(4), 493–500

The Role of Laminar Air Flow and HEPA Filters in Operating Room Ventilation

Laminar air flow plays a fundamental role in reducing the concentration of hazardous substances and controlling airborne contaminants in enclosed environments. In the operating room, the use of this system is of particular importance in directly preventing immediate complications caused by the entry of harmful particles and gases into the bodies of both patients and staff. One of the key components of this system is the HEPA (High Efficiency Particulate Air) filter. These filters are capable of removing more than 99.97% of suspended particles larger than 0.3 microns in diameter. Their precise performance in eliminating microorganisms, dust, and aerosol particles ensures that the operating room environment remains free of microbial and non-microbial contaminants.

The application of laminar air flow in combination with HEPA filters in the operating room not only enhances patient safety and reduces postoperative infections, but also protects operating room personnel against hazards associated with exposure to anesthetic agents, surgical smoke, and other contaminants from an occupational safety perspective.Orthopedic cements—especially when used in large volumes and repeatedly—can pose significant risks to healthcare systems and medical personnel. These materials may contain toxic compounds that can cause adverse effects if inhaled or upon skin contact.

Additionally, there have been reports regarding the potential carcinogenicity of methyl methacrylate (MMA) monomer. When MMA is released and interacts with benzoyl peroxide (BPO), it undergoes oxidation and may convert from a tertiary cyclic amine into an amine and amine oxide compounds known to have carcinogenic potential. However, multiple studies have shown that the tertiary amine often remains chemically unchanged, indicating a need for further investigation into the mechanisms of toxicity and carcinogenesis of these substances.

To reduce the concentration and associated risks of inhaled vapors during the manual mixing of cement, it is recommended to use two benchtop suction units. These devices are designed to capture and remove the gases released during the mixing process, preventing the accumulation of harmful vapors in the environment and enhancing user safety.

Dalhy, M., et al. (2001). Biomaterials, 22(13), 39-47.
Mohan-Kumar. (2020). International Journal

Note

Chapter 5

The Role of Cement Additives

Chapter 5: The Role of Cement Additives

Educational Objectives:

1.To examine the physical and chemical properties of bone cement.
2.To analyze the chemical characteristics of the individual components of cement.

Chemistry and Composition Roles

Powder Components:

1.Role of Polymer (Polymethyl Methacrylate): Forms the primary matrix of the cement.
2.Role of Copolymer: Enhances the physical properties of the cement.
3.Role of Barium Sulfate / Zirconium Dioxide: Provides radiopacity, making the cement visible on radiographic images.
4.Role of Dye (Chlorophyll): Distinguishes cement from bone tissue.
5.Role of Antibiotic (Ab): Prevents microbial infections.

Liquid Components:

1.Role of Monomer (Methyl Methacrylate): Forms the polymer matrix after polymerization.
2.Role of N, N-Dimethyl-p-Toluidine (DMPT): Acts as an initiator for polymerization at low temperatures (cold curing).
3.Role of Benzoyl Peroxide: Reacts with DMPT to catalyze the polymerization process.
4.Role of Hydroquinone: Serves as a stabilizer to prevent premature polymerization.
5.Role of Dye (Chlorophyll): Differentiates cement from bone tissue during revision surgeries.

Mechanical Properties and Tests of Bone Cement

1.Tensile Strength: 25 MPa (Weak)
2.Shear Strength: 40 MPa (Moderate)
3.Young's Modulus of Elasticity: 2004 MPa (Low)
4.Viscoelastic Behavior: Bone cement exhibits viscoelastic properties.

Mechanical Tests:

1. Compression Test:

- Compressive Strength: 70 MPa
- Effect of Antibiotics: The addition of antibiotics does not affect compressive strength.

2. 4-Point Bending Test:

- Effect of Antibiotics on Flexural Strength: The addition of antibiotics reduces flexural strength.

Other Properties:

1. Water Uptake and Saturation:

- Saturation Time: 4 to 8 weeks.

2. Creep:

- Definition: A time-dependent change in strain under constant stress (Lee et al., 2002).
- Function: Reduces internal stresses within the cement and enables partial recovery from plastic deformation.

References:

Lee, A. J. C., et al. (2002). Journal of Biomaterials.

Summary of Properties:

The composition and roles of the components in the powder-polymer system you described can be summarized as follows:

1. Polymer (Polymethyl Methacrylate, PMMA)

- **Role:** Acts as the base material for the cement. PMMA is a biocompatible polymer that provides structural integrity and mechanical strength to the cement.
- **Properties:** Forms a solid matrix upon polymerization, which is essential for the cement's stability and durability.

2. Copolymer

- **Role:** Modifies the physical properties of the PMMA, such as improving flexibility, toughness, or adhesion.
- **Properties:** Enhances the overall performance of the cement, making it more suitable for specific applications (e.g., bone cement in medical applications).

3. Barium Sulfate ($BaSO_4$)

- **Role:** Acts as a radiopacifier.
- **Properties:** Makes the cement visible under X-ray imaging, which is crucial for medical applications to monitor the placement and integrity of the cement in the body.

4. Zirconium Dioxide (ZrO_2)

- **Role:** Also acts as a radiopacifier.
- **Properties:** Similar to barium sulfate, it enhances the visibility of the cement under X-rays. It may also improve the mechanical properties of the cement.

5. Dye (Chlorophyll)

- **Role:** Used to distinguish the cement from surrounding bone or tissue.
- **Properties:** Provides a visual contrast, making it easier for surgeons to identify the cement during and after application.

6. Antibiotic (Ab)

- **Role:** Provides antimicrobial prophylaxis.
- **Properties:** Helps prevent infections by releasing antibiotics locally at the site of application. This is particularly important in orthopedic surgeries to reduce the risk of post-operative infections.

Summary Table:

Component	Role	Properties/Function
Polymethyl Methacrylate	Base polymer	Provides structural integrity and mechanical strength
Copolymer	Modifies physical properties	Enhances flexibility, toughness, or adhesion
Barium Sulfate	Radiopacifier	Makes cement visible under X-rays
Zirconium Dioxide	Radiopacifier	Enhances X-ray visibility and may improve mechanics
Dye (Chlorophyll)	Distinguishes cement from bone	Provides visual contrast
Antibiotic	Antimicrobial prophylaxis	Prevents infections by releasing antibiotics locally

This composition is commonly used in medical applications, such as bone cement in orthopedic surgeries, where radiopacity, antimicrobial properties, and ease of identification are critical.

Liquid Monomer

- **N, N-Dimethyl-p-Toluidine (DMPT):** Acts as an initiator for the cold-curing process of the polymer.
- **Benzoyl Peroxide:** Reacts with DMPT to catalyze the polymerization reaction.
- **Hydroquinone:** Functions as a stabilizer to prevent premature polymerization during storage.
- **Chlorophyll Dye:** Used as a colorant to distinguish bone tissue from cement during revision procedures.

Mechanical Properties and Testing

- **Tensile Strength:** Relatively low, with a maximum of 25 MPa.
- **Shear Strength:** Moderate, reaching up to 40 MPa.
- **Elastic Modulus (Young's Modulus):** Low, approximately 2004 MPa.
- **Material Behavior:** Exhibits viscoelastic properties.

Mechanical Tests:

1. Compression Test: Demonstrates a compressive strength of 70 MPa.
2. 4-Point Bending Test: Evaluates flexural properties.

Effect of Antibiotics:

- The addition of antibiotics does not significantly reduce compressive strength.
- However, antibiotics may decrease bending strength.

Water Uptake and Saturation:

- Water absorption occurs over time, with saturation typically reached between 4 to 8 weeks.

Creep Behavior:

- Creep allows the cement to relax internal stresses and recover from plastic deformation.
- Creep is defined as the time-dependent change in strain under constant stress (Lee et al., 2002).
- In vivo, cement creep can result in multidirectional movement of the cement layer, influenced by temperature and stress conditions (Lee, 2005).

Polymethyl Methacrylate (PMMA)

PMMA is the powder component of bone cement, not the cement itself (cement = polymer + monomer). It is a type of polymer used as a liner or resin in various applications. Its chemical composition and properties are as follows:

Composition and Weight Ratios

- **Powder-to-liquid ratio:** 2:1 (40 g powder to 20 g liquid)

Powder Component:

- 15% PMMA by weight
- 73.5% methyl methacrylate copolymer by weight
- 10% barium sulfate ($BaSO_4$) by weight
- 1.5% benzoyl peroxide (BPO) by weight

Liquid Component:

- 97.4% methyl methacrylate (MMA) by weight
- 75 ± 15 ppm hydroquinone
- 2.6% N, N-dimethyl-p-toluidine (DMPT) by weight

Physical and Chemical Properties

- Colorless, transparent, with a strong odor
- Flammable, supplied in ampoules
- Warning: Using electrocautery devices near this material may cause fire

Biocompatibility and Toxicity

- Chemically highly biocompatible in the body
- Carcinogenicity has not been proven
- May be toxic under certain conditions

Mechanical Properties

- Very hard material, but brittle and fragile

PMMA Polymerization Reactions

PMMA polymerization involves three main stages:

1.Initiation:

- Polymerization starts with initiators like benzoyl peroxide (BPO)
- Activation by N, N-dimethyl-p-toluidine (DMPT)

2.Propagation:

- Growth of polymethyl methacrylate chains

3.Termination:

- Completion of the polymerization reaction

Additives and Their Functions

- **Hydroquinone:** Acts as an inhibitor to prevent unwanted polymerization
- **Barium sulfate ($BaSO_4$) and Zirconia (ZrO_2):** Radiopacifiers to allow X-ray visibility
- **Methyl methacrylate liquid (MMA):** Stabilized with hydroquinone to prevent spontaneous polymerization and maintain its liquid form

PMMA Powder Composition

- Contains initiator (BPO) and radiopaque agents
- Approximately 10% of the powder consists of radiopaque materials for radiographic visibility

Physical and Mechanical Properties of PMMA

Property	Value
Density	1.08 g/cm³
Tensile Strength	25–50 MPa
Compressive Strength	80–100 MPa
Fracture Toughness	1.93 MPa·m^0.5
Fatigue Resistance	1×10^6 to 5×10^6 cycles
Young's Modulus	1500–4000 MPa
Thermal Resistance	100–120 °C
Shrinkage	7%

Mechanical Properties of Bone

Property	Value
Density	2.3 g/cm³
Tensile Strength	2–3 MPa
Compressive Strength	1500 psi (~10.3 MPa)
Shear Strength	1500 psi (~10.3 MPa)
Young's Modulus	10–40 MPa

Sterilization Methods of Orthopedic Cement

Powder Sterilization

- **Gamma Radiation:**
  - Used to sterilize the powder component.
  - May cause polymer chain shortening, potentially altering the mechanical properties of the cement.
- **Ethylene Oxide (EO):**
  - Compared to gamma and beta radiation, EO provides better fatigue performance and causes fewer issues.
  - EO does not significantly affect molecular weight (MW).
  - According to Lewis (2000), cements with higher molecular weight exhibit better fatigue resistance (Tapic, 1996).

Packaging and Sterilization

- **Dental Cement:**
  - Generally, not sterilized.

- **Liquid Component:**
  - Filled into glass ampoules using sterile techniques.
  - Sterilized in blister packaging using EO.
- **Powder Component:**
  - Packaged in plastic or paper containers.
  - Sterilized by gamma radiation or EO.
  - Gamma-sterilized powders require storage for 3 weeks under normal conditions before use (Lewis, 2008).

Re-sterilization of Bone Cement Components

- Re-sterilizing powders previously sterilized by EO using irradiation may cause changes in cement properties, such as alterations in polymer chains.

Summary:

Packing & Sterilization

Dental Cement

- Dental cement is typically not sterilized.

Liquid Component

- The liquid component is filled into glass ampules using sterile techniques.
- It is then packaged in blisters and sterilized using ethylene oxide (EO).

Powder Component

- The powder is packaged in plastic or paper bags.
- It is sterilized using gamma radiation or ethylene oxide (EO).
- Gamma-irradiated powders should be stored for 3 weeks in an ambient environment to ensure proper usability post-sterilization (Lewis 2008).

Re-sterilization of Bone Cement Components

- Re-sterilization of originally EO-sterilized powder using irradiation can alter the properties of the cement (e.g., changes in the polymer chain).

Chemical and Physical Properties of Bone Cement

Powder Component

- **Polymethyl Methacrylate (PMMA):** The main polymer.
- **Copolymer:** Added to modify physical properties.
- **Barium Sulfate ($BaSO_4$) and Zirconium Dioxide (ZrO_2):** Radiopacifiers for X-ray visibility.
- **Dye (e.g., Chlorophyll):** Used to distinguish cement from bone during revision surgeries.

Liquid Component

- **Methyl Methacrylate (MMA) Monomer:** The primary liquid monomer.
- **N, N-Dimethyl-p-Toluidine (DMPT):** Initiator for cold curing polymerization.
- **Benzoyl Peroxide (BPO):** Reacts with DMPT to catalyze polymerization.
- **Hydroquinone:** Stabilizer preventing premature polymerization.
- **Dye (e.g., Chlorophyll):** Used to differentiate cement from bone in revision surgery.

Summary:

Chemical and Physical Properties

Powder Component

- **Polymer Poly Methyl Methacrylate (PMMA):** The primary polymer.
- **Co-polymer:** Added to modify physical properties.
- **Barium Sulfate ($BaSO_4$) and Zirconium Dioxide (ZrO_2):** Used as radio pacifiers for X-ray visibility.
- **Dye (e.g., chlorophyll):** Added to distinguish the cement from bone during revision surgery.

Liquid Component

- **Monomer Methyl Methacrylate (MMA):** The main component.
- **N, N-Dimethyl-p-Toluidine (DMPT):** Initiates cold curing.
- **Benzoyl Peroxide (BPO):** Reacts with DMPT to catalyze polymerization.
- **Hydroquinone:** Acts as a stabilizer to prevent premature polymerization.
- **Dye (e.g., chlorophyll):** Added to distinguish the cement from bone during revision surgery.

Summary and Conclusion:

1.Toxicity of Methyl Methacrylate (MMA):

- MMA may be toxic both locally and systemically.
- There are discussions about MMA and tertiary amines potentially being carcinogenic (Dalhy 2001).
- Although tertiary amines may remain unchanged, their carcinogenicity is not confirmed.

2.Potential Genetic Effects and Persistence in the Body:

- MMA may cause coagulation disorders (hypercoagulation).
- N, N-Dimethyl-p-Toluidine (DMPT) may remain detectable in the body for years.
- Benzoyl Peroxide (BPO) may induce genetic alterations or damage.

3.Biomechanical Properties Compared to Bone:

- The biomechanical properties of bone cement are close to those of bone.

4.Powder Components (At least several substances):

1.Polymethyl Methacrylate (PMMA)
2.Copolymers: methyl methacrylate, ethyl methacrylate, butyl methacrylate
3.Initiators: benzoyl peroxide (BPO), tributyl borane (N-BUMA)
4.Dye: chlorophyll
5.Antibiotics: gentamicin sulfate, clindamycin hydrochloride, erythromycin
6.Radiopacifies: barium sulfate ($BaSO_4$), zirconium dioxide (ZrO_2)

5.Liquid Components (Typically four substances):

1.Methyl methacrylate monomer (MMA)
2.Accelerators: N, N-Dimethyl-p-Toluidine (DMPT)
3.Stabilizer: hydroquinone
4.Crosslinking agents: ethylene glycol dimethacrylate (EGDMA)

Orthopedic Cement Hazards for Personnel (Surgeons and Operating Room Staff)
Health Risks:

1.Allergic Reactions:

- Sensitivity to components of PMMA or antibiotics present in the cement.

2.Skin Complications:
- Contact dermatitis, eczema, redness, swelling, and skin damage due to direct contact with MMA.

3.Eye Complications:
- Eye irritation caused by exposure to vapors or cement particles.

4.Respiratory Complications:
- Shortness of breath (dyspnea), airway spasms, and respiratory system irritation.

5.Systemic Complications:
- Liver dysfunction with long-term exposure.
- Symptoms resembling alcohol poisoning, such as nausea, drowsiness, headache, weakness, fatigue, irritability, confusion, and loss of appetite.

Precautions in the Operating Room and During Work:Fire Hazard:

- **Flammability:**
  - PMMA vapors near electrocautery devices in poorly ventilated environments can pose a fire risk.

Use of Standard Cement:

- **Avoid using:**
  - Cement past its expiration date (usually 6 months).
  - Cement with suspicious packaging or uncertain sterility.

Preparation and Mixing of Cement

- **Homogeneity of Mixture:**
  - Improper mixing of powder and liquid can alter the mechanical and physical properties of the cement.
  - This issue increases the risk of cement mantle fracture or loosening.
- **Exposure to Toxic Vapors:**
  - Odor and Respiratory Effects:
    - Cement emits a strong, pungent odor (similar to spices) during mixing.
    - The tolerable level of MMA for humans is about 125 ppm, but exposure to higher concentrations (e.g., 400 ppm in animals) causes severe respiratory and olfactory irritation.
- **Personal Protection:**
  - **Protective Gear:**
    - Use gloves resistant to MMA penetration; regular latex gloves may

not be sufficient.
- Wear protective clothing and safety goggles to prevent contact with skin and eyes.

- **Special Considerations:**
  - **Pregnancy:**
    - Effects of MMA on pregnant women have not been definitively studied.
  - **Operating Room Ventilation:**
    - Ensure proper ventilation to reduce the concentration of toxic vapors in the environment.

Special Notes on Sensitivity and Adverse Effects for Personnel

- **Contact Dermatitis:**
  - **Causes:**
    - Direct contact with cement during mixing, adding to the prosthesis, or fixing the prosthesis.

Prevention:

- Use double gloves and replace gloves promptly.
- Avoid latex or perforated gloves as they may worsen symptoms.
- Use protective goggles (shield) to prevent cement particles from contacting the eyes, which is essential for the surgical team.

Fire Hazard (Inflammability)

- **Toxic Vapors:**
  - Vapors from the monomer (MMA) and cement mixing have a strong, pungent odor that can irritate the respiratory tract, eyes, and skin.
  - In sensitive personnel, these vapors may cause airway spasms or exacerbate asthma.
- **Fire Risk:**
  - Using electrocautery (electrosurgical device) near the cement, especially in poorly ventilated environments or in the presence of oxygen sources, can increase the risk of fire.

Hypersensitivity Reactions

- **Causing Agents:**
  - The monomer (MMA), cement powder, factory-added components, or antibiotics present in the cement may trigger hypersensitivity reactions.

- **Symptoms:**
  - Severe allergic reactions such as anaphylaxis can occur in personnel or patients.

Animal Studies and Cement Complications

- **Skin and Systemic Effects:**
  - Repeated contact with cement may cause dermatitis, numbness, burning sensations, and tingling.
- **Respiratory Complications:**
  - Inhalation of MMA vapors in laboratory animals has irritated the respiratory and olfactory systems.
- **Effects on Pregnant Women:**
  - Animal studies have shown that exposure to MMA may affect the fetus.
  - Pregnant women are advised to avoid inhaling MMA vapors.

Measures to Reduce Complications

1.Use of Appropriate Ventilation Systems:

- Installation of Laminar Air Flow systems to reduce the concentration of toxic vapors in the environment (Vertical HEPA System).

2.Isolation of Cement Mixing Area:

- Creating a separate environment with glass walls for cement mixing to minimize staff and patient exposure to toxic vapors.

3.Use of Protective Equipment:

- Wearing resistant gloves, protective goggles, and appropriate respiratory masks.

4.Staff Training:

- Educating personnel about the hazards of cement and safe handling practices.

Comprehensive References for Further Study:

1. Schummer, W. et al. (2014). Bone cement embolism attached to the Central Venous catheter. British Journal of Anesthesia, 112, 672-674.

2.Scoinick, B. & Collins, J. (1986). Systemic reaction to methyl methacrylate in an operating room nurse. Journal of Occupational Medicine, 28, 196-198.
3.British Journal of Anesthesia (2015). Hazards of bone cement for the patient and operating theater personnel. Volume 114, Issue 1, 168-169.
4.Leggat, P.A. et al. (2009). Surgical application of methyl methacrylate. Occupational Medicine, 64, 207-212.

Properties of PMMA (Polymethyl Methacrylate) and related notes:

1.General properties of PMMA:

- It is lightweight: its density ranges from 1.20 to 1.70 g/cm³, which is about half the density of glass.
- Impact resistance: it has good impact strength.
- Elastic modulus: the elastic modulus of PMMA lies between cancellous bone and cortical bone:
  1.Cortical bone
  2.PMMA
  3.Cancellous bone
- Melting point: melts at 46°C (860°F).
- Behavior when burning: when burned, it produces carbon dioxide, water, carbon monoxide, and formaldehyde.
- Transparency: transmits 92% of light at a thickness of 3 mm.
- Water absorption: PMMA has low water absorption (0.3 to 0.4%).

2.Exothermic reactions:

- In laboratory conditions, PMMA polymerization produces heat up to 80°C.
- In the body, this heat reaches about 40 to 50°C.
- Copolymers such as butyl acrylate are used to improve impact strength.
- Methacrylic acid is used to improve the glass transition temperature.
- Plasticizers are used to improve plastic properties and impact resistance.

3.Physical properties and composition of PMMA:

- Composition: PMMA is mixed as 2 parts powder and 1 part liquid (MMA monomer).
  - Liquid (MMA monomer): colorless and the source of cement odor.
  - It is 97-99% liquid (Jaeblon, 2010).
  - Contains N, N-dimethyl-p-toluidine as a reaction accelerator.
  - Contains hydroquinone as a monomer stabilizer.
- Product differences: the main difference between various PMMA prod-

ucts lies in their powder composition.

- PMMA powder contains radiopaque materials such as barium sulfate ($BaSO_4$) and zirconia (ZrO_2).
- Initiators: such as benzoyl peroxide or tri-n-butyl borane.
- Accelerators: such as 2,5-dimethylhexane and hydroperoxide.
- Dyes: such as chlorophenyl in Palacos R monomer and ethanol and ascorbic acid in CM monomer (Depuy).

Thermal Necrosis:

- The heat generated during the polymerization reaction of PMMA can cause necrosis of the surrounding tissues. This phenomenon is especially important in orthopedic applications such as arthroplasty.

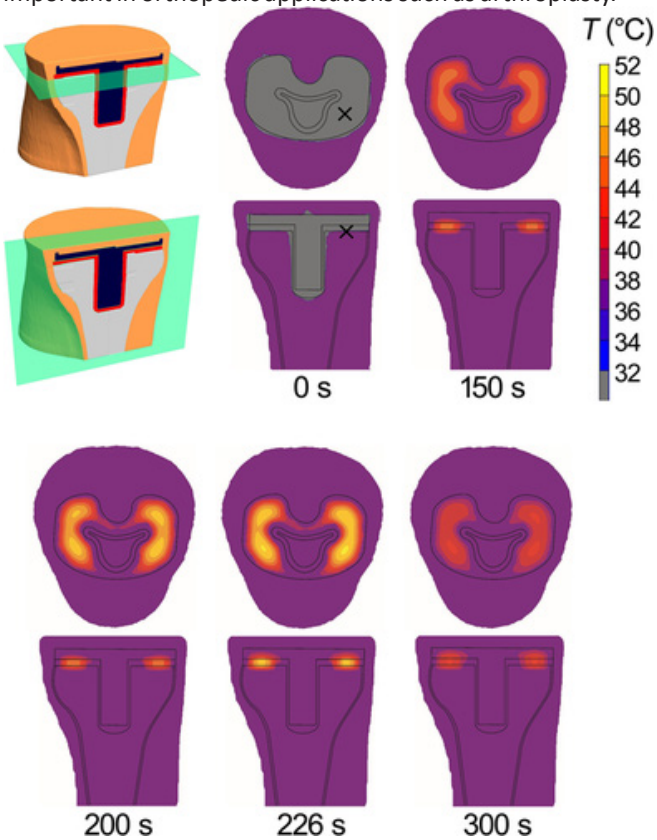


Fig 9. Thermal Necrosis Caused by Bone Cement

Comprehensive References for Further Study:

1. Lewis, G. (2008). Properties of acrylic bone cement: State of the art review. Journal of Biomedical Materials Research Part B: Applied Biomaterials, 86(2), 569-573.
2. Jaeblon, T. (2010). Polymethylmethacrylate: Properties and contemporary uses in orthopaedics. Journal of the American Academy of Orthopaedic Surgeons, 18(5), 297-305

Thermal Necrosis of Bone Cement

Heat Generation:

- The polymerization of PMMA is an exothermic reaction that generates considerable heat.
- The surrounding tissues absorb this heat and may lead to thermal necrosis.

Critical Temperature and Exposure Time:

- Exposure of tissue to 50°C for 1 minute or 47°C for 5 minutes can result in bone necrosis.
- Temperatures between 47°C and 60°C cause osteocyte damage, activation of the NF-kappa-B
- pathway, and increased expression of cytokines (such as M-CSF) and prostaglandins (via COX-2 enzyme).
- These processes interfere with the differentiation of osteoclasts and osteoblasts (Erikson, 1983).

Factors Influencing Thermal Necrosis:

1.Type of Cement:

- The chemical composition and viscosity of the cement affect the amount of heat generated.

2.Cement Mantle Thickness:

- A mantle thickness greater than 5–7 mm increases the risk of thermal necrosis.

3.Size and Material of the Implant:

- Larger or metallic implants may absorb more heat.

4.Type of Fixation Technique:

- Cemented fixation of the acetabular component may increase the risk of thermal necrosis (Huiskes, 1980).

Factors Related to Cement Polymerization

- Mixing Technique:

Different mixing methods (e.g., manual technique) affect heat distribution and polymerization rate.

- Ambient Temperature:

The room temperature influences the reaction rate and heat generation.

- Cement Viscosity:

High-viscosity cements may produce more heat and increase the risk of bone necrosis (Gergely, 2016).

Combination of Powder and Liquid (PMMA + MMA)

- **Combination Process:**

Mixing PMMA powder with MMA liquid creates an exothermic chemical reaction.

Thermal analyses indicate that temperatures at the cement-bone interface can reach 100–145°C.

- **In Vivo Studies:**

Animal studies have shown that the heat generated during polymerization can cause bone necrosis and collagen damage.

Conclusion:

- Thermal necrosis is a serious complication of bone cement use, resulting from the exothermic polymerization of PMMA.
- Several factors, such as cement type, cement mantle thickness, ambient temperature, and mixing method, influence the severity of this complication.
- Employing appropriate mixing techniques, controlling ambient temperature, and monitoring cement thickness can reduce the risk of thermal necrosis.

Summary:

Thermal Necrosis and Bone Cement Polymerization

Exothermic Reaction and Heat Generation

- The polymerization of PMMA (polymethyl methacrylate) bone cement is an exothermic reaction, generating significant heat.
- Critical temperatures and exposure durations for thermal bone necrosis:
  - 50°C for 1 minute or 47°C for 5 minutes can induce bone necrosis.
  - Temperatures between 47–60°C damage osteocytes, activate inflammatory pathways (e.g., NF-kappa-B, M-CSF, COX-2), and disrupt osteo-

blast/osteoclast differentiation (Erikson, 1983).

Factors Influencing Thermal Necrosis

1.Cement Mantle Thickness:

- Mantles thicker than 5–7 mm increase heat retention and necrosis risk.

2.Implant Characteristics:

- Larger or metallic implants may absorb and retain more heat.

3.Cement Type and Viscosity:

- High-viscosity cements generate more heat during polymerization (Gergely, 2016).

4.Mixing Technique and Room Temperature:

- Manual mixing methods (e.g., Montel technique) and ambient temperature affect reaction speed and heat distribution (Huiskes, 1980).

Clinical Implications

- **Cement-Bone Interface:**
  - In vivo studies report polymerization temperatures reaching 100–145°C at the interface, causing collagen denaturation and osteocyte death.
- **Acetabular Component Fixation:**
  - Cemented acetabular components pose higher necrosis risk due to thicker cement layers.

Mitigation Strategies

- Control Cement Mantle Thickness: Limit to <5 mm where possible.
- Optimize Mixing Techniques: Use vacuum mixing to reduce porosity and heat generation.
- Monitor Ambient Temperature: Cooler environments slow polymerization, reducing peak temperatures.

Key References

Erikson, A. P. (1983): Linked thermal damage to inflammatory mediator release.
Huiskes, R. (1980): Highlighted the role of cement thickness and mixing methods.
Gergely, R. C. (2016): Demonstrated risks of high-viscosity cements.

Chapter 6

Complications of Bone Cement During Surgery

Chapter 6: Complications of Bone Cement During Surgery

Bone Cement Implementation Syndrome (BCIS)

Bone Cement Implementation Syndrome (BCIS) is a potentially life-threatening complication that typically occurs during or immediately after orthopedic surgeries involving the use of bone cement, such as total hip or knee arthroplasty. The syndrome results from the entry of bone cement materials, fat, or other embolic particles into the bloodstream, which can lead to serious outcomes such as hypoxia, hypotension, cardiac arrhythmias, and even death.

Signs and Symptoms of BCIS:

1. Mild form:

- Hypoxemia (reduced blood oxygen saturation)
- Confusion (altered mental status)
- Fat embolism-like symptoms (including respiratory and neurological issues)

2. Severe form:

- Severe hypotension (drop in blood pressure)
- Cardiac arrhythmias (irregular heart rhythm)
- Respiratory failure
- Seizures
- Cardiac arrest
- Mortality

Diagnostic Challenges:

- Diagnosing BCIS can be difficult, as its clinical presentation may resemble other intraoperative complications such as fat embolism syndrome or allergic reactions.
- Symptoms can progress rapidly, requiring prompt identification and intervention.

Risk Factors:

- Use of bone cement in orthopedic surgeries
- Major procedures such as total hip replacement

- Elderly patients or those with pre-existing cardiovascular or pulmonary conditions

Prevention and Management:

- Close monitoring of the patient's vital signs during and after surgery
- Employing optimal surgical techniques to minimize intravascular embolization of cement or debris
- Immediate treatment of hypoxia and hypotension with supplemental oxygen and appropriate pharmacologic support

Reference:
Singla et al., "Paravara Syndrome," Med Rev 2014; 6(2):

Acute/Intraoperative Complications

1.Cardiovascular:

- Bone Cement Implantation Syndrome (BCIS): A critical syndrome linked to PMMA, characterized by:
  - Hypotension, arrhythmias, ECG changes.
  - Hypoxia ($\downarrow O_2$ saturation).
  - Increased pulmonary vascular resistance (PVR, likely typo for "PUR").
- Severe events: Cardiac arrest, myocardial infarction, pulmonary embolism, heart block.

2.Central Nervous System (CNS):

- Confusion, loss of consciousness (especially during tourniquet deflation or cement injection).
- Cerebral vascular accident (CVA/stroke).

3.Peripheral Neurovascular Damage:

- Direct trauma or embolism causing nerve/vascular occlusion.

4.Life-Threatening Risks:

- Sudden death, often due to BCIS or massive embolism.

5.Transient Effects:

- Temporary hypotension, thrombophlebitis, hematoma, hemorrhage, fever, hematuria, dysuria.

Delayed Postoperative Complications

1.Local Surgical Complications:

- Prosthetic component loosening.
- Deep wound infection (e.g., biofilm formation, noted as "G-Bio film").
- Trochanteric bursitis (in THA), heterotopic ossification (abnormal bone growth).
- Trochanteric separation (hip hardware failure).

2.Neurovascular/Visceral Damage:

- Local neuropathy, bladder fistula (possibly a typo for "bowel" or "vesical" fistula).
- Intestinal obstruction due to implant extrusion.

Key Mechanisms

- BCIS: Hypotension and hypoxia result from PMMA monomer toxicity, embolization of marrow contents, or immune-mediated reactions during cement hardening.
- Tourniquet Deflation: Sudden release of emboli into circulation.
- Delayed Infections: Biofilm formation on implants (e.g., Staphylococcus spp.).

Clinical Implications

- **Preventive Measures:**
  - Slow cement injection, adequate hydration, and hemodynamic monitoring to mitigate BCIS.

- Antibiotic-loaded cement to reduce infection risk.

- **Monitoring:** Vigilance for embolic events during cementation and tourniquet release.

Severity of Bone Cement Implantation Syndrome (BCIS):

BCIS is clinically classified into three grades based on the severity of symptoms:

Grade I – Mild

- **Moderate hypoxia:**
  - Oxygen saturation (SpO_2) < 94%
- **Mild hypotension:**
  - Systolic blood pressure drop > 20% from baseline
- **Prevalence:** ~20% of cases

Grade II – Moderate

- **Severe hypoxia:**
  - SpO_2 < 88%
- **Severe hypotension:**
  - Systolic BP drop > 40% from baseline
- **Altered consciousness:**
  - Sudden loss of consciousness or confusion
- **Prevalence:** ~3% of cases

Grade III – Severe

- **Cardiopulmonary collapse:**
  - Acute heart or lung failure
  - Requires cardiopulmonary resuscitation (CPR)
- **Prevalence:** ~1% of cases

Summary of Classification:

Grade	Clinical Criteria	Prevalence
I	SpO_2 < 94%, SBP ↓ > 20%	~20%
II	SpO_2 < 88% or SBP ↓ > 40% + decreased consciousness	~3%
III	Cardiopulmonary collapse requiring CPR	~1%

This grading system helps with the rapid identification and appropriate management of BCIS.

Incidence and Clinical Features of Bone Cement Implantation Syndrome (BCIS):

Incidence of BCIS:

- The precise incidence of BCIS is not independently established, as the syndrome is often grouped under general cardiopulmonary events.
- To investigate BCIS, clinicians must examine intraoperative or postoperative cardiopulmonary complications, especially in cemented orthopedic procedures (e.g., using PMMA).
- **Important note:**
  - In scientific research, PMMA (Polymethylmethacrylate) should not be confused with Para-methoxy methylamphetamine, a psychoactive substance.
  - Due to a lack of clear diagnostic criteria, there are no accurate mortality statistics for BCIS. Sudden deaths on the operating table after cement insertion should be considered probable BCIS.

Clinical Features of BCIS:

Life-Threatening Signs Following Cementation:

1.Hypoxemia:
- Decreased oxygen saturation (SpO_2).

2.Life-threatening Hypotension:
- Significant drop in systolic blood pressure.

3.Pulmonary Hypertension:
- Due to fat embolism or increased pulmonary vascular resistance.

4.Cardiac Dysrhythmia:
- Arrhythmias caused by hypoxia or PMMA monomer toxicity.

5.Cardiac Shock:
- Acute cardiac failure.

6.Pulmonary Edema:
- Secondary to pulmonary hypertension or cardiac failure.

7.Bronchoconstriction:
- Airway narrowing due to inflammatory reactions.

8.Cardiac Arrest:
- In severe cases.

9.Sudden Death:
- Due to fat embolism or cardiopulmonary collapse.

Additional Signs:

- **Decrease in $EtCO_2$ (End-tidal CO_2):**
  - Indicates reduced cardiac output or pulmonary embolism.
- **Increase in CVP (Central Venous Pressure):**
  - It occurs due to increased pulmonary artery pressure or right heart failure.
- **Hypothermia:**
  - It occurs as a result of systemic reactions to fat embolism.
- **Thrombocytopenia:**
  - It occurs due to activation of the coagulation system and consumption of platelets.

Summary:

- The incidence of BCIS is not precisely defined, and it should be evaluated among perioperative cardiopulmonary events in orthopedic surgery.
- Key clinical signs include hypoxemia, hypotension, dysrhythmia, cardiac shock, pulmonary edema, and sudden death.
- Rapid recognition and prompt management of these symptoms are essential to prevent life-threatening complications.

Etiology, Pathophysiology, and Management of Bone Cement Implantation Syndrome (BCIS):

Etiology of BCIS:

Causes of Embolism:

1.Gas (Air):

- Originating from methyl methacrylate (MMA).

2.Fat:

- Entry of fat into the bloodstream.

3.Debris:

- Entry of particulate matter into circulation.

Aggravating Factors:

1.Increased Intramedullary Pressure:

- Especially observed in the femoral stem area during total hip arthroplasty (THA).

2.Vasodilatory Effects of Cement:

- Induced by bone cement.

3.Toxic Effects:

- Due to the chemical components of the cement.

Models of BCIS:

Although a combination of mechanisms leads to specific clinical symptoms, two primary explanatory models exist:

1. Embolic Model:

- **Diagnosis:**
  - Identified through right atrial and left ventricular echocardiography and transesophageal echocardiography (TE echo).
- **Postmortem Confirmation:**
  - Reported in autopsy findings.

2. Mediator Model:

- **Vasoconstriction:**
  - Resulting from the release of inflammatory mediators such as histamine.
- **Causes:**
  - Hypersensitivity reactions.
  - Complement activation via C3 and C5.

Prevention, Treatment, and Management of BCIS:

1.Close Monitoring of Patients:

- Especially essential for those with cardiovascular comorbidities.
- Early detection and hemodynamic/respiratory support yield favorable outcomes.
- Invasive monitoring is crucial in cases involving hip fractures and long-stem implants.

Etiological Considerations in BCIS:

Although the exact etiology of BCIS is not fully elucidated, key factors include:

1. Methyl Methacrylate (MMA) Monomer:

- Causes vasodilation.
- This effect has not been observed in vivo during arthroplasty.

2. Embolization:

- Movement of debris such as bone marrow, cement particles, fat, air, bone fragments, platelets, and fibrin into the right atrium.
- Embolization typically occurs in the right ventricle and is subsequently transmitted to the pulmonary vessels.

Embolization Steps:

1.Femoral reaming
2.Acetabular cementing
3.Femoral cementing
4.Prosthesis insertion
5.Joint reduction

- Using standard techniques, 93.4% of embolic events occur during prosthesis insertion.
- With modified techniques, this rate reduces to 13.4%.

Mediator Theory:

1.Increased Pulmonary Vascular Resistance (PVR):

- Follows the release of inflammatory mediators.
- Results from ventilation-perfusion mismatch.

2.Complement Activation:

- C3a and C5a mediators contribute to pulmonary vasoconstriction.
- Elevated histamine levels have been reported in some patients.

3.Transesophageal Echocardiography (TE Echo):

- Shows significant differences in embolic cascade between cemented and uncemented prostheses.

Predisposing Patient Characteristics:

1.Age > 65 years.

2.Twice as common in women as in men.

3.More prevalent in patients with higher ASA classifications (ASA 3, 4) and uncontrolled comorbidities.

Additional Notes:

Percutaneous Kyphoplasty and Vertebroplasty (PVP):

- First reported by Galibert in 1987.
- Similar BCIS-like complications have been observed.

Bone Cement Embolism (BCE) Complications:

Cause of BCIS:

- Occurs during the cold curing phase, when large volumes of cement particles enter the bloodstream.
- Presents with clinical symptoms such as hypoxia and hypotension, potentially fatal in severe cases.

Clinical Case Example:

Case Description:

- A 65-year-old female undergoing kyphoplasty for kyphosis received 3 mm of cement into the thoracic vertebra.
- The patient experienced ventricular tachycardia (VT) during surgery.
- Initial management with lidocaine, amiodarone, and magnesium sulfate was ineffective.
- ABGs, blood chemistry, and cardiac enzymes were within normal ranges.
- Transthoracic echocardiography (TTE) in the recovery room revealed a foreign body (likely bone cement) in the right ventricle.
- A CT scan was recommended by the radiologist.

Statistics:

- Cement leakage or embolism occurs in 4.6% of kyphoplasty cases.

Treatment:

1.Supportive Therapy:

- For pulmonary embolism: oxygenation and hemodynamic stabilization.

2.Surgical or Percutaneous Embolectomy:

- Recommended for peripheral embolisms.

Reference:

Jonathan B. Chon, MD. The American Society of Anesthesiology. In: Lippincott Williams & Wilkins, Anesthesiology 2012; 117(2): 4-7.

Grades of Bone Cement Implantation Syndrome (BCIS):

Grade 1 (G1):

- **Moderate Hypoxia:**
  - Oxygen saturation (SpO_2) less than 94%.
- **Hypotension:**
  - More than 20% decrease in Systolic Arterial Pressure (SAP).

Grade 2 (G2):

- **Severe Hypoxia:**
  - Oxygen saturation (SpO_2) less than 88%.
- **Severe Hypotension:**
  - More than 40% decrease in Systolic Arterial Pressure (SAP).

Grade 3 (G3):

- **Cardiovascular Collapse:**
  - Requires Cardiopulmonary Resuscitation (CPR).
  - Sudden reduction in left ventricular volume and cardiac output.
- **Diagnostic Studies:**
  - Transcranial Doppler studies show evidence of emboli in cerebral circulation.

High-Risk Groups for Developing BCIS:

1. Patient-Related Factors:

- **High ASA Class (ASA III, IV):**
  - Patients with controlled comorbidities (e.g., diabetes or renal failure).
- **Pulmonary Hypertension:**
  - Increased pulmonary artery pressure exacerbates embolic risk.
- **Osteoporosis:**
  - Weakened bone structure may necessitate greater cement volume.

2. Surgery-Related Factors:

- **Pathologic Fracture:**
  - Due to underlying conditions such as bone tumors.
- **Hip Fracture:**
  - Especially in elderly patients with low bone density.
- **Arthroplasty with Long Stem Prosthesis:**
  - Use of longer prostheses may facilitate entry of cement particles into the bloodstream.

Reference:

Edmonds, C. R., Barbut, D., Hager, D., & Sharrock, N. E. (2000). Intraoperative cerebral arterial embolization during total hip arthroplasty. Anesthesiology, 93, 315-318.

Summary:

Prevention of Bone Cement Implantation Syndrome (BCIS):

1. Preoperative Assessment and Optimization

- **Risk Evaluation:** Conducted by an anesthetist to identify high-risk patients (e.g., those with cardiac/pulmonary comorbidities, osteoporosis, or limited cardiopulmonary reserve).
- **Comorbidity Management:** Optimize cardiovascular health (e.g., control hypertension, manage heart failure) and ensure stable hemodynamics before surgery.

2. Intraoperative Monitoring

- **Hemodynamic Monitoring:** Utilize cardiac output (CO) monitoring, transesophageal echocardiography (TEE), and Doppler to detect emboli and assess cardiac function.
- **Pulmonary Artery Catheterization:** Monitor pulmonary artery pressures to identify early signs of embolism or right heart strain.

3.Surgical Techniques for Prevention

- **Canal Preparation:**
  - **Thorough Lavage:** Use pulsed lavage to remove bone debris, fat, and marrow from the intramedullary canal, reducing embolic load.
  - **Debris Removal:** Clear residual material to minimize embolism risk during cement insertion.
- **Cement Application Strategies:**
  - **Retrograde Insertion:** Apply cement from distal to proximal to allow air/marrow, reducing intramedullary pressure.
  - **Vacuum Cementing:** Use vacuum systems to evacuate air and reduce embolus formation.
  - **Low-Viscosity Cement:** Employed with controlled pressure to minimize intramedullary pressure spikes (note: technique-specific; some protocols may prefer high-viscosity cement depending on surgeon preference).
- **Hemostasis:** Maintain surgical field clarity to facilitate effective lavage and cement application.

4. Additional Considerations

- **Venting/Drainage:** Create vents or use drains in the femur to relieve pressure during cementation.
- **Cement Timing:** Allow cement to reach optimal viscosity (avoiding overly liquid states) to reduce leakage risk.
- **Alternative Prostheses:** Consider uncemented implants for high-risk patients when feasible.

Conclusion: A multimodal approach combining preoperative optimization, vigilant intraoperative monitoring, and meticulous surgical technique is critical to mitigating BCIS risk. Tailor strategies to patient-specific factors and surgical context.

Prevention of BCIS

1.Preoperative assessment of patients (Anesthetist) (risk and consult for comorbidity assessment)
2.Optimize the cardiovascular system
3.Use of intraoperative CO monitoring (Semi TE echo) & Doppler monitor & pulmonary artery flotation catheter
4.Prevention:

- Thorough lavage of the intramedullary canal
- Removal of debris
- Good hemostasis
- Bone vacuum cementing
- Retrograde insertion of cement (distal to proximal)
- Use low–viscosity cement

Key Considerations for the Prevention of Bone Cement Implantation Syndrome (BCIS) and Anesthesia Selection:

1. Choice of Anesthesia Method (General vs. Spinal):

- **No proven superiority between techniques:**

There is no conclusive evidence that general anesthesia is superior to spinal anesthesia or vice versa. The choice of anesthetic technique should be based on patient-specific factors, clinical status, and the preferences of both the surgeon and anesthesiologist.

- **High cardiovascular risk patients:**

In patients with higher ASA classification (III or IV), uncontrolled hypertension, or hip fractures, comprehensive cardiovascular assessment and consultation with relevant specialists are essential. In such cases, spinal anesthesia may be preferred due to its potential to reduce cardiovascular and respiratory complications.

2. Techniques to Reduce BCIS Risk:

Vacuum Mixing of Cement:

- **Reduction of vasoactive release:**

Mixing the cement powder and liquid under vacuum reduces the release of vaso-

active substances (e.g., methyl methacrylate), which may lower the risk of BCIS.

- **Improved strength and homogeneity:**

This method also enhances the mechanical strength and homogeneity of the cement, potentially decreasing mechanical complications such as cement fracture.

3. General Considerations for High-Risk Patients:

- **Comprehensive preoperative assessment:**

A full cardiac and respiratory evaluation is mandatory before surgery in high-risk patients.

- **Optimization of patient status:**

Controlling blood pressure, improving cardiopulmonary function, and managing comorbidities can help reduce the risk of BCIS.

- **Intraoperative monitoring:**

Utilizing advanced monitoring techniques such as transesophageal echocardiography and hemodynamic monitoring enables early detection and management of complications.

Summary:

For high cardiovascular risk patients, the choice of anesthesia should be individualized, weighing the advantages and disadvantages of each method. Advanced techniques such as vacuum mixing of cement can reduce the incidence of BCIS and improve cement quality. Thorough preoperative assessment and vigilant intraoperative monitoring are crucial to minimizing complications.

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Summary:

Management of Bone Cement Implantation Syndrome (BCIS): Key Principles

1.Early Diagnosis and Suspicion:

- BCIS is reversible if diagnosed and managed promptly. A high index of suspicion is critical, especially in high-risk patients undergoing cemented arthroplasty.

2.Aggressive Resuscitation and Supportive Care:

- Immediate and aggressive management is essential to reduce morbidity and mortality associated with BCIS.

3.Key Management Strategies:

- **Oxygenation:** Maintain oxygen saturation at 100% using supplemental oxygen or mechanical ventilation if necessary.
- **Inotropic Therapy:** Administer inotropes (e.g., epinephrine, norepinephrine) to support blood pressure and cardiac output.
- **Cardiac Output (CO) Monitoring:** Use advanced monitoring (e.g., arterial line, transesophageal echocardiography) to assess and optimize hemodynamics.
- **Central Venous Pressure (CVP) Line:** Insert a CVP line to guide fluid therapy and inotropic drug administration.
- **ICU Care:** Transfer the patient to the intensive care unit (ICU) for close monitoring and stabilization of cardiopulmonary function.

4.Supportive Measures:

- **Fluid Resuscitation:** Administer intravenous fluids cautiously to maintain adequate preload without exacerbating right heart strain.
- **Ventilatory Support:** Use mechanical ventilation if respiratory failure occurs.
- **Management of Arrhythmias:** Treat arrhythmias promptly with antiarrhythmic medications or cardioversion as needed.
- **Monitoring for Complications:** Watch for signs of pulmonary embolism, right heart failure, or disseminated intravascular coagulation (DIC).

5.Prognosis:

- With early diagnosis and aggressive management, BCIS is reversible, and patients can recover fully. Delayed treatment can lead to significant morbidity and mortality.

Conclusion:

The management of BCIS requires a high level of suspicion, early diagnosis, and aggressive supportive care. Oxygenation, inotropic support, hemodynamic monitoring, and ICU care are critical components of treatment. Prompt intervention can significantly improve outcomes and reduce the risk of complications.

Importance of Early Diagnosis and Intervention in BCIS (Bone Cement Implantation Syndrome):

1.Early Diagnosis:

- A high index of suspicion for BCIS is crucial, as it necessitates immediate medical intervention.

2.Goal of Fluid Therapy:

- To reduce pulmonary arterial pressure (PAP) and improve right ventricular function.

3.Administration of Potent Inotropes:

- Medications such as dobutamine and milrinone are essential to maintain left ventricular performance.

4.Cardiac Function Monitoring:

- Cardiac performance should be closely monitored using transesophageal echocardiography (TEE) and the PiCCO (Pulse Contour Cardiac Output) system.

5.High-Risk Patients:

- In high-risk individuals, especially those with hip fractures, routine cardiac monitoring is necessary to detect pulmonary artery clotting.

Summary of Bone Cement Implantation Syndrome (BCIS):

1.Definition:

BCIS is characterized by sudden physiological instability, including:

- Hypoxia
- Hypotension
- Arrhythmias
- Increased pulmonary vascular resistance (PVRT)

2.Timing:

- Occurs intraoperatively, typically 5–160 seconds after cementation.

3.Incidence:

- Not rare but a serious complication.
- Between 2005–2012, 2,900 cases were reported by the National Reporting and Learning System (NRLS) in England and Wales, often associated with cardiac arrest or peri-arrest situations.

4.Etiology:

- Embolization of air, gas, fat, bone debris, or methyl methacrylate (MMA) monomers.
- Hypersensitivity reactions to cement components.

5.Patient Risk Factors:

- Comorbidities: Pulmonary hypertension, cardiac disease (ASA III-IV).
- Pathologic fractures or hip fractures.
- Long-stem prostheses (higher risk than short-stem THA).

6.Anesthetic Considerations:

- Hemodynamic instability due to inadequate monitoring or poor preoperative optimization.

7.Prevention:

- Identify high-risk patients: Those with cardiac or pulmonary disease, advanced age, male sex, or on diuretics.
- Use of checklists and adherence to NICE Guidelines (2015).
- Intraoperative measures: Thorough lavage, vacuum cementing, retrograde cement insertion, and hemodynamic monitoring.

8.Management:

- Early recognition of symptoms (e.g., hypoxia, hypotension).
- Physiological support: Oxygenation, inotropes, and fluid resuscitation.
- Hemodynamic monitoring: Use of arterial lines, CVP, and cardiac output monitoring.

Conclusion:

BCIS is a potentially life-threatening complication of cemented arthroplasty. Prevention through patient identification, meticulous surgical technique, and adherence to guidelines is critical. Early recognition and aggressive physiological support are essential to reduce morbidity and mortality.

Adverse Reactions to Bone Cement:

1.Bone Cement Implantation Syndrome (BCIS):

- Transitory fall in blood pressure (BP)
- Cerebrovascular accident (CVA)
- Cardiovascular complications:
  - Pulmonary embolism (PE)
  - Myocardial infarction (MI)

2.Biochemical Complications:

Elevated serum levels of gamma-glutamyl transpeptidase (GGTP) for up to 10 days postoperatively.

3.Mechanical Complications:

- Trochanter separation
- Infection (superficial and deep)
- Trochanteric bursitis
- Long-term complications:
  - Osteolysis
  - Aseptic loosening
  - Fragmentation or breakage of the cement mantle
  - Wear

4.Inflammatory and Tissue Reactions:

- Local inflammation and necrosis
- Foreign body reaction

Causes of Adverse Reactions:

1.Absorption of Methyl Methacrylate (MMA):

- Systemic absorption of MMA by the body.

2.Bone Marrow Embolization:

- Entry of bone marrow particles into the bloodstream.

3.Conversion of MMA to Methacrylic Acid:

- Metabolic conversion of MMA to methacrylic acid within the body.

(Reference: Joint Replacement 2017)

Allergic Reactions to Orthopedic Cement:

- Allergic reactions to orthopedic cement are rare.
- In some cases, hypersensitivity may occur to specific components of the cement.
- In dentistry, patients may report a history of allergic reactions to similar materials (Thomas et al., 2008b).
- MMA is not detectable in serum during the mixing of orthopedic cement (Homlar, 2013).

Chapter 7

Standardization and Bone Cement

Chapter 7: Standardization and Bone Cement

1.Standardization of Bone Cement (PMMA) by FDA:

- **Approval History:**
  Orthopedic bone cement (PMMA), after over 30 years of research and review, received market authorization in the United States for use in arthroplasty surgeries.
- **Class II (Special Controls):**
  The cement was ultimately classified as a Class II medical device, requiring special controls and strict quality monitoring.

2.Acceptance and Update Process:

- **1993:**

PMMA cement was initially classified as Class III (requiring Premarket Approval).

- **2001:**

With updated standards, the product received a 510(k)-clearance indicating compliance with safety and effectiveness requirements. This update upgraded the cement to Class II, facilitating market access.

3.Importance of Standardization:

- **Safety and Effectiveness:**
  FDA standardization ensures bone cements are safe and effective for clinical use (2001).
- **Quality Control:**
  Manufacturing, storage, and application processes are strictly monitored to prevent adverse effects.

Table of Risks Associated with Orthopedic Bone Cement Use (FDA):

1. Bone Cement Implantation Syndrome (BCIS)
2. Polymerization and Setting Problems
3. Loosening or Migration of the Prosthesis
4. Infection and Fever
5. Adverse Tissue Reaction
6. Pain and Loss of Function
7. Need for Revision Surgery

Evaluations Performed on Orthopedic Bone Cement:

1.Biocompatibility Tests:

Necessary tests to ensure compatibility with body tissues.

2.Physical and Chemical Property Assessment:

Thorough evaluation of physical and chemical characteristics.

3.Mechanical Tests and Validated Documentation:

- Modulus of elasticity
- Fatigue resistance
- Fracture toughness
- Fatigue crack
- Flexural strength
- Compressive strength
- Shearing strength
- Tensile strength
- Creep behavior

Bone Cement Preparation Techniques:

1.Operating Room Temperature Impact:

The mixing process is dependent on OR ambient temperature.

2.Centrifugation:

Speed, time, and volume are precisely evaluated for optimization.

3.Modeling:

Design, materials, and temperature considerations are included.

4.Sterilization Recommendations:

Use of dry heat, ethylene oxide (ETO), or radiation methods is advised.

Mixing Methods:

- **Hard Mixing:**
  Likely manual mixing involving physical force.
- **Mechanical Mixing:**
  Use of mechanical devices to ensure uniform mixing.
- **Vacuum Mixing:**
  Mixing under vacuum to reduce air bubbles, enhancing cement strength.

Contraindications:

- Presence of active or untreated infection in the targeted area (localized).

Precautions:

1.Patient Monitoring:

- Continuous blood pressure monitoring (hypotension occurs 10 to 165 seconds after cement application and risk persists up to 50 minutes afterward).
- Assessment of cardiovascular signs (such as arrhythmia, myocardial infarction, or stroke).

2.Thermal complications:

- Cement temperature may reach 55°C, causing tissue burns (exothermic reaction).

3.Avoidance of excessive pressure:

- Excessive pressure may cause damage to adjacent tissues or extrusion of cement from the site.

4.Special populations:

- Use is contraindicated in children and pregnant women (safety not established).
- Do not use expired cement.

5.Dermatologic complications:

- Possible contact dermatitis in medical personnel.

Side effects and risks:

- **Cardiovascular:**
  - Hypotension, myocardial infarction (MI), cerebrovascular accident (CVA), pulmonary embolism, sudden death.
- **Hemorrhage and thrombosis:**
  - Hematuria, thrombophlebitis, local bleeding.

- **Mechanical:**
  - Prosthesis loosening, trochanteric bursitis, trochanter displacement, micromotion (micro-movement at bone-cement interface).
- **Rare:**
  - Heterotopic ossification (H.O.), delayed sciatic nerve injury.
- **Others:**
  - Fever with dysuria, allergic reactions.

Key points to reduce risks:

- Close patient monitoring during the first hour after cement application.
- Use of standardized techniques to minimize heat generation.
- Avoid use in patients with history of hypersensitivity or uncontrolled severe heart failure.
- Check cement expiration date and storage conditions.

Effect of cement viscosity based on brand and composition:

Bone cement viscosity varies depending on brand and composition, which affects its application and performance. Appropriate viscosity provides the surgeon sufficient time for cement handling, molding, and precise prosthesis placement. Additionally, viscosity influences cement penetration into cancellous bone tissue and the formation of strong physical bonds.

Important characteristics of cement viscosity:

1.Opportunity for prosthesis molding and placement:

A. Appropriate viscosity allows the surgeon enough time to convert cement from liquid to dough state and position the prosthesis accurately. This is particularly important in total hip arthroplasty (THA) and use of cement gun.

B. Low viscosity cement has a short waiting time and hardens rapidly. This is advantageous in spine surgeries (vertebroplasty and kyphoplasty) where rapid setting is desired but offers limited molding time.

2.Penetration into trabecular bone and hemorrhage control:

A. Suitable viscosity allows cement to penetrate deeply into trabecular bone and form strong physical bonds.

B. Low viscosity cement, due to its runny state, penetrates trabecular bone better but may offer less resistance to bleeding.

C. High viscosity cement, due to its dough state, has better hemorrhage resistance but lower penetration into trabecular bone.

Types of cement viscosity:

1.Low Viscosity:

- Characteristics:
  - Runny state.
  - Short waiting time with rapid hardening
  - Deep penetration into trabecular bone.
- Advantages:
  - Suitable for surgeries requiring deep cement penetration (e.g., vertebroplasty).
- Disadvantages:
  - Limited time for molding and prosthesis placement (In my opinion, it is preferable in TKA, stem THA, and spine procedures).
  - Low resistance to bleeding.

2.Medium Viscosity:

- Characteristics:
  - Dual phase.
  - Intermediate between low and high viscosity.
- Advantages:
  - Balance between trabecular bone penetration and hemorrhage resistance.
  - Adequate time for molding and prosthesis placement.
- Applications:
  - Suitable for most orthopedic surgeries, especially total hip arthroplasty (THA) and total knee arthroplasty (TKA).

3.High Viscosity:

- Characteristics:
  - Dough state.
  - Ready for use immediately after mixing.
- Advantages:
  - High resistance to bleeding.
  - Adequate time for molding and prosthesis placement.
- Disadvantages:
  - Low penetration into trabecular bone.
  - Higher risk of thermal necrosis due to exothermic reaction.

- More reports of early failure in total knee arthroplasty (TKA) (Jansen 2012).

Key points:

- Low viscosity:

Excellent penetration into trabecular bone.
Short waiting time and limited molding opportunity.

- Medium viscosity:

Balance between penetration and hemorrhage resistance.
Suitable for most orthopedic surgeries.

- High viscosity:

High hemorrhage resistance.
Higher risk of thermal necrosis and early failure.

Summary:

- Selection of appropriate viscosity depends on surgery type, patient bone condition, and surgeon preference.
- Medium viscosity is suitable for most surgeries due to its balanced properties.
- Low viscosity is useful for surgeries requiring deep penetration (e.g., vertebroplasty).
- High viscosity may carry higher risks in total knee arthroplasty (TKA).
- Some orthopedic cements use terms such as Standard Viscosity or medium viscosity in their product descriptions. However, the fundamental problem is that, in many cases, no precise and standardized definition has been provided for this classification. Therefore, it is unclear whether these types of cements should be categorized as Low Viscosity, Medium Viscosity, or High Viscosity in practice.

Basic recommendations

for the use of cement in orthopedic surgeries, especially joint prostheses such as knee or hip:

1.Appropriate cement selection:

- Medium or high viscosity cement: Due to better rheological properties, these cements have greater permeability into cancellous bone tissue, leading to stronger prosthesis-bone bonding and enhanced prosthesis stability.
- High viscosity cements also provide longer working time, allowing more precise placement by the surgeon.

2.Preferred mixing method:

- Use of optimal mixing technique: Mixing method plays a critical role in final cement quality. Vacuum mixing devices reduce air bubbles in cement and increase density and strength.
- Precise timing: Mixing and application timing must be carefully controlled to ensure proper polymerization.

3.Bone preparation:

- Bone lavage: Before cement application, bone must be thoroughly washed to remove fat, blood, or soft tissue that may interfere with cement bonding.
- Cement pressurization: After cement placement, pressurizing cement into cancellous bone tissue (using syringes or specialized tools) can improve penetration and bonding.
- Cement should be applied at the appropriate phase (usually when viscosity is increased but not fully hardened).

4.Environmental temperature control:

- Proper temperature: Ambient and cement temperatures should be controlled to prevent overly rapid or slow polymerization. High temperatures cause rapid hardening; low temperatures prolong the setting time.

5.Postoperative patient monitoring:

- Regular follow-up: After surgery, the patient should be regularly monitored to ensure the absence of complications such as prosthesis loosening or infection.

Summary:

Viscosity Types of Bone Cement and Their Impact on Handling and Fixation

The viscosity of bone cement significantly influences its handling characteristics, penetration into bone, and the overall quality and longevity of the fixation. Below is a detailed breakdown of the different viscosity types and their effects:

1. Low Viscosity Cement

- **Characteristics:**
  - Short waiting phase (no dough state).
  - Short true working time (can be picked up with a gloved hand quickly).
  - Variable setting time.
  - Runny state, which allows for deep penetration into cancellous bone (spongy bone).
- **Advantages:**
  - Excellent penetration into cancellous bone due to its liquid-like consistency.
- **Disadvantages:**
  - Short working time requires a quick application.
  - May be less resistant to bleeding at the implantation site.
  - Higher risk of micro-motion during early fixation due to its runny state.
- **Clinical Considerations:**
  - Suitable for procedures where deep cement penetration is critical.
  - Requires careful monitoring due to rapid setting.

2. Medium Viscosity Cement

- **Characteristics:**
  - Intermediate viscosity (dual-phase cement).
  - Versatile and suitable for most procedures.
  - Balanced handling time and working phase.
- **Advantages:**
  - Offers a balance between ease of handling and penetration into bone.
  - Less runny than low-viscosity cement, reducing the risk of micro-motion.
  - Resistant to bleeding during application.
- **Clinical Considerations:**
  - Ideal for a wide range of orthopedic procedures, including total hip arthroplasty (THA) and total knee arthroplasty (TKA).
  - Provides reliable fixation with moderate working time.

3. High Viscosity Cement

- **Characteristics:**
  - Ready to apply immediately after mixing (no waiting phase).
  - Stiff consistency, making it difficult to interlock with spongy bone.
  - Short working time, requiring close monitoring.
- **Advantages:**
  - Reduced risk of cement run-off during application.
  - Better resistance to bleeding.
  - Predictable handling due to its dough-like consistency.
- **Disadvantages:**
  - Higher risk of early failure due to poor interdigitation with cancellous bone.
  - Associated with micro-fractures and thermal necrosis in THA (Jansen 2012).
  - May not penetrate deeply into bone, leading to weaker fixation.
- **Clinical Considerations:**
  - Suitable for procedures where rapid fixation is needed.
  - Requires careful application to avoid thermal damage and micro-fractures.

Comparison of Viscosity Types

Property	High Viscosity	Medium Viscosity	Low Viscosity
Waiting Phase	None (ready to apply immediately)	Short to moderate	Long
Working Time	Short	Intermediate	Short
Penetration into Bone	Poor (too stiff for deep penetration)	Moderate	Excellent (deep penetration)
Resistance to Bleeding	High	Moderate	Low
Handling Characteristics	Stiff, dough-like	Balanced	Runny, liquid-like
Risk of Early Failure	Higher	Lower	Moderate
Thermal Necrosis Risk	Higher (due to exothermic reaction)	Moderate	Lower

Key Clinical Implications

1.Handling Time:

- Low viscosity: Short working time1, quick setting.
- Medium viscosity: Balanced working time, versatile.

- High viscosity: Immediate application2, short working time.

2.Penetration into Cancellous Bone:

- Low viscosity: Deep penetration, ideal for weak or osteoporotic bone.
- High viscosity: Poor penetration, may lead to weaker fixation.

3.Resistance to Bleeding:

- High viscosity: Best resistance.
- Low viscosity: Poor resistance, may require additional measures.

4.Risk of Thermal Necrosis:

- High viscosity: Higher risk due to exothermic reaction.
- Low viscosity: Lower risk, but still possible.

5.Longevity of Fixation:

- Medium viscosity: Optimal balance for most procedures.
- High viscosity: Higher risk of early failure due to poor interdigitation.

Regarding loosening and the need for revision arthroplasty, no significant difference has been observed between high-viscosity and low-viscosity cements (Kalasan et al., Journal of Arthroplasty, 2020).

High-viscosity cement in vertebroplasty demonstrates lower leakage and may provide better pain relief, while low-viscosity cement allows for deeper bone penetration and better anatomical correction.

Feature	Low Viscosity (LV)	High Viscosity (HV)
Liquid phase	Long-lasting liquid phase	Very short phase
Working time	Short working phase	Very short; rapid loss of stickiness
Initial adhesion (String time)	High, due to deep penetration	Lower, due to limited penetration
Surgeon handling time	Longer working phase for the surgeon	Shorter phase, requiring quick decision-making
Bone penetration	Deep cancellous bone penetration	Limited penetration; potential for weak bonding
Resistance to fat and bone debris	Prevents the ingress of fat and bone particles into cement	Low resistance to particles and blood (may need additional measures)

Article References:

- Spender (2007)
- Hass (1975)
- Klein (2004)
- Hazelwood (2015, 2019)
- Jansen (2012)

Cemented THA (Technique in THA):

In Total Hip Arthroplasty (THA) using bone cement, two general cementing techniques are commonly employed: the Traditional and the Modern methods. The modern method involves advanced steps aimed at enhancing cement penetration into the bone and creating microscopic interlocking to improve mechanical stability, prosthesis fixation, and reduce long-term complications such as prosthesis loosening. The two techniques are detailed below:

1. Traditional Cementing Technique:

- This is the older approach and involves simpler cementing steps.
- **General steps:**
  - Bone preparation using manual tools.
  - Use of cement without pressurization.
  - Implantation of the prosthesis into the cement before complete hardening (in the doughy phase, when it does not stick to gloves).
- **Disadvantages:**
  - Lower cement penetration into cancellous bone.
  - Potential formation of air bubbles in the cement, which reduces strength.
  - Higher long-term risk of prosthesis loosening.

2. Modern Cementing Technique:

This method includes advanced steps that significantly improve the quality of cementing. The key stages include:

A. Plugging the Femoral Canal

- Before cement placement, the femoral canal is occluded with a plug (typically plastic or cement) at least 2 cm distal to the tip of the prosthesis.
- This step ensures pressurization of the cement within the femoral canal and improves penetration into the cancellous bone.

B. Pulsating Lavage

- High-pressure pulsatile lavage is used to wash the femoral and acetabular bone surfaces.
- This removes blood, fat, and bone debris, preparing the surface for better cement adhesion.

C. Removal of Soft or Hard Cancellous Bone

- Any soft or sclerotic cancellous bone that may prevent cement penetration is removed using specific instruments.
- This creates an optimal surface for deeper cement interdigitation into the bone.

D. Pressurization

- After cement placement in the femoral canal, pressurizers are used to force the cement under pressure into the cancellous bone.
- This enhances cement-bone adhesion and reduces the risk of prosthesis loosening.

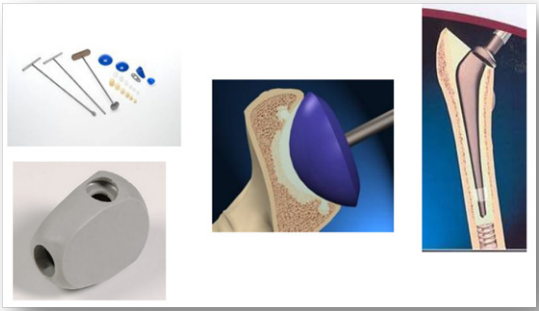


Fig 10. Pressurization of Bone Cement

E. Porosity Reduction

- Vacuum mixing systems are used to eliminate air bubbles from the cement.
- Reduced porosity increases the strength, durability, and homogeneity of the cement.

F. Centralization of the Prosthesis

- Centralizers are employed to position the prosthesis centrally within the femoral canal.
- This promotes uniform cement distribution around the prosthesis and reduces the risk of loosening.

Advantages of the Modern Technique:

- Improved cement penetration into cancellous bone.
- Reduced cement porosity and increased mechanical strength.
- Better prosthesis stability and lower risk of long-term loosening.
- Reduced infection risk due to improved bone decontamination.

Conclusion:

The modern cementing technique in THA provides superior outcomes com-

pared to the traditional method due to its advanced procedures and technologies. It is particularly recommended for patients with low bone density (osteoporosis) or in cases requiring long-term prosthesis stability.

Technical Considerations in the Application of Orthopedic Bone Cement:

Technical precision in the application of orthopedic bone cement is crucial, as the quality of bone-cement bonding directly impacts the stability and longevity of the prosthesis (Bone-Cement Micro-Interlock). The following are detailed key points:

1. Bone Surface Preparation:

- **Removal of Debris:** All bone fragments, soft tissue, debris, fat, and other contaminants must be thoroughly removed from the bone surface. This improves the cement's adhesion to the bone.
- **Drying the Bone Surface:** The bone surface must be completely dry to ensure optimal cement bonding. The presence of moisture or blood can significantly reduce adhesion.
- **Removal of Synovium and Soft Tissue:** In the distal femur area, synovium and soft tissues around the bone cuts should be excised using electrocautery or a surgical blade to ensure direct contact between the cement and bone.

2. Use of Pulse Lavage and Brush:

- **Pulse Lavage:** The use of high-pressure pulsatile irrigation helps remove blood, fat, and debris from the bone surface. This enhances cement penetration into the cancellous bone and reduces the risk of prosthesis loosening.
- **Brush:** The use of a bone brush to clean the surface can further improve cement adhesion by ensuring a cleaner and more receptive bone interface.

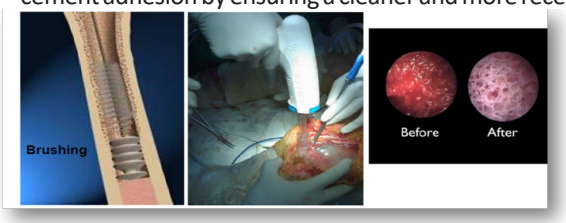


Fig 11. Pulse Lavage

A high-pressure fluid irrigation system thoroughly cleans the surgical site

and removes debris, improving surface preparation. Brush
Used to mechanically scrub and debride bone surfaces, enhancing cleanliness and promoting better adhesion of implants or bone cement.

3.Creating Small Holes in Sclerotic Surfaces:

- On sclerotic surfaces of the tibia or acetabulum, small holes (3–4 mm in diameter) can be drilled using a Low-numbered drill bit (not a hammer) to improve cement bonding. This enhances cement penetration and strengthens micro-interlocking.
- This method can increase bone–cement bonding strength by approximately 8–18%.

4. Cement Extrusion from the Greater Trochanter Area (Swearing Trochanter Sign):

- Occasionally, cement may extrude from the greater trochanter area, a phenomenon known as the Swearing Trochanter Sign. This typically occurs due to excessive cement pressurization and requires attention to prevent complications.

5. Precautions During Cement Mixing:

- **Avoid Electrocautery Use:** Electrocautery must not be used during cement mixing or when in contact with monomer, as methyl methacrylate is highly flammable and may ignite in the presence of oxygen.
- **Protection of Skin and Mucosa:** The monomer is a strong lipid solvent and must not come into contact with skin, mucous membranes (especially the respiratory tract), or eyes. Surgical personnel should wear appropriate PPE.

6. Control of Mixing Speed:

- **Mixing Speed:** Aggressive or high-speed mixing accelerates polymerization and reduces the setting time, which may complicate cement handling and placement.
- **Powder-to-Liquid Ratio:** The standard ratio is 2:1 (powder to liquid). Altering this ratio affects the setting time:
  - Increased liquid → prolonged setting time.
  - Decreased liquid → shorter setting time.
  - Increased powder → shorter setting time.
  - Decreased powder → prolonged setting time.

7. Reducing Cement Porosity:

- The use of vacuum mixing devices can effectively remove air bubbles from the cement and reduce porosity, thereby improving its strength and durability.

Conclusion:

Adhering to these technical steps during cement application improves bone–cement micro-interlock, enhances prosthesis stability, and reduces long-term complications such as prosthetic loosening. Careful attention to detail is essential for optimal outcomes.

Use of Vacuum System:

The use of vacuum systems for mixing and applying orthopedic cement offers considerable advantages that improve cement quality and extend implant longevity. The following highlights the benefits of this system:

1. Reduction of Micro-Porosity:

- **Air Bubble Elimination:** Vacuum systems remove air bubbles from the cement, thereby reducing porosity. Air bubbles act as weak points and compromise mechanical strength.
- **Enhanced Cement Strength:** Reduced porosity increases cement density and mechanical strength, improving prosthesis stability.

2. Better Homogeneity of the Mixture:

- **Uniform Mixing:** Vacuum systems allow for homogeneous mixing of powder and liquid, preventing the formation of clumps.
- **Improved Mechanical Properties:** Homogeneity improves mechanical properties and contributes to long-term implant success.

3. Extended Implant Longevity:

- **Lower Risk of Loosening:** Reduced porosity and improved cement properties lower the risk of prosthetic loosening over time.
- **Stronger Fixation:** Higher-quality cement creates a stronger bond between the implant and bone.

4. Vertical vs. Horizontal Vacuum Systems:

- **Vertical Type:** Vertically designed vacuum systems are preferred, as air bubbles escape more efficiently.
- **Horizontal Type:** Less effective; air bubbles may remain trapped, compromising cement quality.

5. Optimal Vacuum Pressure:

- **0.25 to 0.5 Bar:** This range is considered optimal, effectively eliminating air bubbles without causing phase separation (powder/liquid).
- **Excessive Pressure:** May lead to cement phase separation and re-

duced quality.

6. Reduced Working Time:

- **Faster Working Window:** Vacuum systems may shorten cement handling time, allowing the surgeon to apply the cement at the optimal moment with improved precision.

Conclusion:

The vacuum system significantly enhances orthopedic cement quality by reducing porosity, improving homogeneity, increasing implant longevity, and lowering the risk of loosening. Vertical vacuum systems operating at 0.25–0.5 bar are considered optimal. Adherence to these practices enhances surgical outcomes and patient satisfaction.

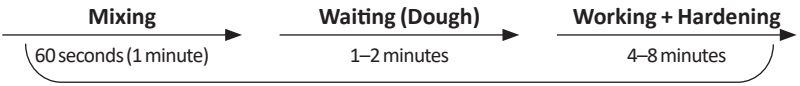
Example – Palacos® R+G Fast:

- With specialized formulation and vacuum mixing, the powder-to-liquid ratio and polymerization rate are adjusted to achieve both dough time and setting time under 5 minutes at room temperature.
- The addition of gentamicin in the powder provides localized antibacterial action while maintaining mechanical strength.
- These features make Palacos® Fast ideal for CVP techniques and small joint prostheses, allowing for accurate and efficient cement application with reduced infection risk.

Bone Cement (B.C) Processing and Application

The process of bone cement (B.C) handling and application involves several stages, each of which holds significant chemical and physical importance. These stages are generally divided into four main phases: Mixing, Waiting, Working, and Hardening.

The following is a detailed explanation of these phases and the chemical and physical events occurring during the process:



1.Mixing Phase:

- **Duration:** Approximately 60 seconds.
- **Process:** The liquid, usually methyl methacrylate (MMA), is added to the cement powder and thoroughly mixed to create a paste with a consistency similar to toothpaste.
- **Chemical and Physical Events:**
  1.Dissolution of powder in liquid (MMA): The cement powder, typically polymethyl methacrylate (PMMA), dissolves in the MMA liquid.
  2.Initiation of polymerization: The chemical polymerization reaction begins, where MMA monomers link together, forming polymer chains.
  3.Increase in viscosity: As polymerization progresses, the cement's viscosity gradually increases.

2.Waiting Phase:

- **Duration:** From the end of mixing until the start of the working phase.
- **Process:** The cement reaches a sticky dough state and becomes ready for application.
- **Chemical and Physical Events:**
  - Polymerization continues, and viscosity further increases.
  - Cement changes from a paste to a sticky state.

3.Working Phase:

- **Duration:** Approximately 60 seconds to 4–8 minutes (depending on cement type and environmental conditions).
- **Process:** The cement reaches a dough-like, semi-solid state, ready for placement into the bone.
- **Chemical and Physical Events:**

- Polymerization continues, and the cement gradually hardens.
- Cement viscosity reaches a level suitable for easy implantation into the bone.

4.Hardening Phase:

- **Duration:** Approximately 2–3 minutes.
- **Process:** Cement fully hardens and becomes solid.
- **Chemical and Physical Events:**
  - Polymerization reaction completes, and cement attains its final state.
  - Exothermic reaction: This stage releases heat, raising cement temperature to 70–80°C.
  - Cement fully hardens, ready for prosthesis fixation.

Key Times in Cementing Process:

- Dough Time: Total time from mixing start until the cement becomes non-sticky. Typically, 2–3 minutes.
- Working Time: Total time from mixing start until cement becomes too stiff to work with. Typically, 8–10 minutes.
- Setting Time: Total time from mixing start until cement is fully hardened. Typically, 8–13 minutes.

Chemical and Physical Events During Mixing:

- Chemical:
  1.Dissolution of PMMA powder in MMA liquid.
  2.Initiation of polymerization as MMA monomers link to form polymer chains.
- Physical:
  - Increase in cement viscosity due to ongoing polymerization.

Conclusion:

The processing and application of bone cement involve complex stages with significant chemical and physical changes. Understanding these phases and key time points assists surgeons in effectively and safely applying cement, thereby enhancing prosthesis stability. Strict adherence to these stages can improve surgical outcomes and reduce long-term complications.

Bone Cement Polymerization

Polymerization is the process by which liquid monomers, typically methyl

methacrylate (MMA), convert into solid polymers. This process plays a key role in bone cement, leading to its hardening and the development of mechanical strength. Below are the details of this process and its effects:

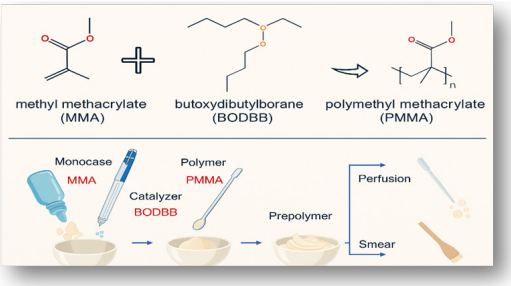


Fig. 12. Polymerization is the process by which liquid monomers, typically methyl methacrylate (MMA), are converted into solid polymers.

Polymerization Stages

1.Breaking Carbon-Carbon Double Bonds:

- At the initiation of the process, an initiator (usually benzoyl peroxide) is activated and generates free radicals.
- These free radicals attack the carbon-carbon double bonds (C=C) in the MMA monomers, causing these bonds to break.

2.Formation of Polymer Chains:

- MMA monomers link together, forming long polymer chains.
- This chain reaction continues, converting the liquid monomer into a solid polymer.

3.Cross-linking:

- Polymer chains bond with each other, creating a three-dimensional (3D) network.
- This 3D network provides bone cement with mechanical strength and resistance.

Polymerization Characteristics

1.Exothermic Reaction:

- Polymerization is an exothermic reaction, meaning it produces heat.
- In laboratory conditions, the temperature can reach 75 to 100 °C (167 to 212 °F).
- In the body, due to heat absorption by surrounding tissues, the temperature is usually limited to 40 to 50 °C.

2.Increase in Viscosity:

- As polymerization progresses, the cement viscosity increases, transitioning from liquid to dough-like and finally solid.
- This viscosity increase allows the surgeon time to shape the cement and position the prosthesis accurately, especially in CVP at the acetabulum.

3.Polymerization Time:

- The polymerization time depends on various factors such as temperature, cement type, and monomer-to-polymer ratio.
- Typically, polymerization in the body completes within 10 to 15 minutes.

Effects of Polymerization in the Body

1.Heat Generation:

- The heat produced during polymerization may cause thermal necrosis in adjacent tissues.
- This effect is more pronounced in high-viscosity cements.

2.Strength Formation:

- Polymerization forms strong physical bonds between the cement and bone.
- These bonds aid in prosthesis stabilization and increase its longevity.

3.Bleeding Control:

- The increase in viscosity and polymer network formation reduces bleeding at the surgical site.
- In TKA, cement is used to fill the reamed areas on both sides of the tibial and femoral medullary instead of grafts.

Key Points

- Polymerization is an exothermic process that produces heat.
- In the body, temperature is limited to 40–50 °C, whereas it can reach 100 °C in laboratory settings.
- The increase in viscosity during polymerization provides the surgeon with time for shaping and implant placement.
- The heat generated may cause thermal necrosis in adjacent tissues, especially with high-viscosity cements.

Two Mixing Phases in Bone Cement

Mixing bone cement is a critical step that affects the mechanical proper-

ties and final performance of the cement. Two main mixing methods are depending on the cement type, such as liquid-to-powder (e.g., CMW) and powder-to-liquid (e.g.,Palacos-R). The details of these methods and their impacts are as follows:

1.Mixing Method in CMW Cement: Liquid Added to Powder

- Mixing Process:
  - In this method, the liquid monomer, usually methyl methacrylate (MMA), is added to the polymer powder (polymethyl methacrylate, PMMA).
  - Due to the low viscosity of the monomer, this method allows for uniform mixing.
- Advantages:
  - Fast and homogeneous mixing.
  - Reduced risk of air bubble entrapment in the cement.
- **Disadvantages:**
  - Manual mixing may reduce the mechanical properties of the cement.

2.Mixing Method in Palacos-R Cement: Powder Added to Liquid

- Mixing Process:
  - In this method, the polymer powder is added to the liquid monomer.
  - Due to the higher viscosity, this method requires more careful mixing.
- Advantages:
  - Better control over cement viscosity.
  - Reduced likelihood of air infiltration.
- Disadvantages:
  - Manual mixing may increase cement porosity.

Effect of Manual Mixing on Cement Properties

- **Reduced Mechanical Properties:**
  - Manual mixing can introduce air bubbles, increasing cement porosity.
  - Porosity may reach up to 6.6%, which reduces mechanical strength.
- **Increased Porosity:**
  - Excessive stirring increases porosity and decreases cement quality.
  - To avoid this, mixing should be performed at a frequency of 1–2 Hz for 1–2 minutes.

Polymerization Process and Role of Initiators

- **Radical Generation:**
  - Initiators such as benzoyl peroxide (BPO) in the powder and activators like DMPT (dimethyl-p-toluidine) in the liquid start the polymerization process (DMP + BPO).
    These generate free radicals that attack the carbon-carbon double bonds (C=C) in the monomers, initiating polymerization.
- **Exothermic Reaction:**
  - Polymerization is exothermic and generates heat.
  - This heat can cause thermal necrosis in adjacent tissues.

Effect of Heat on Prosthesis Loosening

- **Thermal Necrosis:**
  - Sudden temperature rises during cement setting may cause thermal necrosis in surrounding tissues.
  - This can contribute to prosthesis loosening.
- **Scientific Studies:**
  - Studies such as Huiskes (1980) and MJoberg (1986) indicate that sudden heat increase may cause loosening, though this remains debated.
  - Some research suggests that appropriate techniques (e.g., vacuum mixing) can reduce risks of thermal necrosis and loosening.

Summary of Mixing

- Bone cement mixing must be performed carefully to minimize porosity and maintain mechanical strength.
- Manual mixing may increase porosity and reduce cement strength.
- Polymerization is exothermic and can cause thermal necrosis and prosthesis loosening.
- Advanced techniques like vacuum mixing improve cement quality.

Volume Shrinkage Phenomenon and Its Effects

Volume shrinkage is a major challenge in bone cement use, caused by physical and chemical changes during polymerization, potentially affecting cement performance and surgical outcomes. Details are as follows:

1.Volume Shrinkage of Cement

- **Mechanism:**
  - During polymerization, liquid monomers (e.g., MMA) convert to solid polymers.
  - This leads to a volume reduction of about 6–7%.
  - Volume loss occurs due to density changes before and after polymerization.
- **Reported Values:**
  - For simple PMMA cements: ~1.5% shrinkage.
  - For antibiotic-loaded cements: 2.5–2.8% shrinkage (Teuber 2010).
- **Clinical Impact:**
  - Shrinkage can create gaps between cement and bone or prosthesis.
  - These gaps may lead to aseptic loosening.

2.Factors Affecting Cement Shrinkage

- **Mixing Method:**
  - Hand Mixing:
  Introduces air, increases porosity (up to 6.6%), exacerbating shrinkage.
  - Gun Mixing or Vacuum Centrifugation:
  Reduce air entry and porosity, minimizing shrinkage.
- **Cement Viscosity:**
  - High viscosity cements are more prone to shrinkage.
  - Use of cement guns is recommended for high viscosity cements.
- **Polymerization Speed:**
  - Fast polymerization increases heat and shrinkage.

- Slow polymerization minimizes shrinkage.

3.Paradoxical Behavior of Cement

- **Slight Shrinkage:**
  - Cement loses some volume during polymerization.
- **Slight Swelling:**
  - In some cases, slight swelling occurs post-polymerization.
- **Cause:**
  - Due to fluid absorption or pressure changes in the environment.

4.Curing Process

- **Definition:**
  - Curing is the phase where cement reaches its final state.
- **Effect of Heat:**
  - Higher heat intensity accelerates curing speed.
  - Excess heat may cause thermal necrosis in surrounding tissues.
- **Cause of Heat:**
  - The liquid monomer part of cement is responsible for heat generation (Kuhn 2000, 2001).

5.Clinical Recommendations

- Use advanced mixing techniques such as vacuum mixing or cement gun injection to reduce porosity and shrinkage.
- Prefer medium viscosity cements to lower prosthesis loosening risk.
- Avoid titanium resin or its alloys in revision arthroplasty (Willert et al., 1996).

Summary

- Volume shrinkage is common in bone cements and may lead to aseptic loosening.
- Advanced mixing methods and appropriate cement selection can minimize this phenomenon.
- The paradoxical volume behavior (shrinkage and slight swelling) and curing are important factors to consider.

Biomechanical Behavior of Bone Cement

Orthopedic bone cement, primarily composed of polymethyl methacrylate (PMMA), serves as a critical biomechanical material in orthopedic surgeries, playing a vital role in prosthesis fixation. Understanding the physiological response of the body to bone cement and its biomechanical properties—especially the phenomenon of creep—is essential to improve implant performance and longevity. The following discussion outlines these aspects:

Physiological Response of Bone Cement in the Body

Bone cement generally acts as an inert material, but it elicits specific physiological responses:

1. **Transient Inflammatory Reaction:** After implantation, a localized inflammatory response occurs in the surrounding tissues, which usually subsides over time.
2. **Formation of a Fibrous Layer:** Over time, a fibrous membrane forms around the cement, effectively isolating it from adjacent tissues.
3. **Lack of Biological Bonding:**
   PMMA cement does not biologically bond with bone. Instead, it establishes a mechanical interlock (micro-interlock), contributing to the mechanical stability of the implant.

Biomechanical Properties of Bone Cement

Bone cement exhibits distinct biomechanical properties that determine its clinical behavior:

1. Creep (Time-Dependent Deformation)

- **Definition:** Creep refers to the slow, permanent deformation of a solid material under sustained mechanical stress. In bone cement, this occurs due to compressive and shear forces.
- **Relevance in Orthopedics:**
  - **Low Creep is Beneficial:** In the early postoperative period, a small degree of creep aids in stress distribution and reduces localized pressure around the prosthesis.
  - **Excessive Creep is Detrimental:** High levels of creep can lead to cement deformation and compromise prosthetic stability.
- **Factors Influencing Creep:**
  - **Porosity:** Cement with higher porosity exhibits lower resistance to creep.
  - **Viscosity:** Higher-viscosity cement usually has greater resistance to creep.
  - **Time:** The creep rate decreases over time as the cement continues to harden.

2. Compressive and Tensile Strength

- Bone cement must possess sufficient compressive and tensile strength

to endure mechanical loads transmitted through the prosthesis.

3. Elastic Modulus

- The elastic modulus of the cement should be compatible with that of the surrounding bone to minimize stress shielding, which can lead to bone resorption.

4. Adhesion

- The cement must adhere well to both the bone and prosthesis surfaces to ensure a stable interface. This is enhanced by bone surface preparation (e.g., periosteal scraping and removal of debris).

5.Impact of Porosity and Viscosity on Biomechanical Properties

- **Higher Porosity:** Leads to reduced mechanical strength and lower creep resistance.
- **Higher Viscosity:** Enhances the cement's resistance to creep and deformation, improving fixation integrity.

Conclusion

Bone cement functions as a biomechanical anchor in joint arthroplasty. The creep behavior of cement is a double-edged sword—minimal creep postoperatively can help with load distribution, while excessive creep compromises prosthesis stability. Porosity and viscosity are two crucial factors influencing creep resistance. A thorough understanding and appropriate control of these properties are essential for improving prosthetic longevity and clinical outcomes.

Summary:

"What is the physiological response of bone cement in the body, and what biomechanical properties does it induce?"

In materials science, creep (sometimes referred to as cold flow) is the tendency of a solid material to slowly deform or flow permanently when subjected to mechanical stresses, even if those stresses are below the material's yield strength. This phenomenon typically occurs over long periods of time and is especially significant at elevated temperatures, though it can also happen at room temperature in certain materials (referred to as cold creep).

Key Points:

1.Mechanical Stress: Creep occurs when a material is under constant stress, such as tension, compression, or shear.

2.Time-Dependent Deformation: Unlike elastic deformation (which is instantaneous and reversible), creep happens gradually over time and is often irreversible.

3.Temperature Dependency: Creep is more pronounced at high temperatures, but some materials exhibit creep even at room temperature.

Desirable vs. Undesirable Creep:

- **Desirable Creep:** In some applications, a small amount of creep can be beneficial. For example:
  - In tightening bolts, a small amount of creep can help maintain tension over time.
  - In polymer components, creep can allow for stress relaxation, which can be useful in sealing applications.
- **Undesirable Creep:** Excessive creep is generally harmful because it can lead to:
  - **Deformation:** Parts may change shape, leading to failure or malfunction.
  - **Fracture:** Over time, creep can cause materials to crack or break.
  - **Reduced Lifespan:** Components may fail prematurely due to accumulated deformation.

Examples of Creep in Real Life:

1.Turbine Blades in Jet Engines: These operate at high temperatures and stresses, making them susceptible to creep. Over time, the blades can de-

form, affecting engine performance.

2.Concrete Structures: Concrete can experience creep under constant load, leading to sagging or deformation over decades.

3.Polymers and Plastics: Materials like PVC pipes can slowly deform under constant pressure, especially at elevated temperatures.

Stages of Creep:

1.Primary Creep: Rapid initial deformation that slows down over time.

2.Secondary Creep: Steady-state deformation at a constant rate.

3.Tertiary Creep: Accelerated deformation leading to failure.

In summary, creep is a critical consideration in materials science and engineering, particularly for components exposed to high stresses and temperatures. While a small amount of creep can be beneficial in some cases, excessive creep is generally undesirable and must be managed through proper material selection and design.

Porosity of Bone Cement

Porosity refers to the proportion of void spaces between solid and rigid material particles, which can potentially be filled with other substances. An increase in porosity within bone cement leads to a reduction in mechanical strength, a decrease in fatigue resistance, and an overall increase in material vulnerability. This phenomenon primarily results from air trapping during the wetting and mixing stages of cement preparation. Elevated porosity may contribute to aseptic loosening of the prosthesis and a reduced service life of implants.

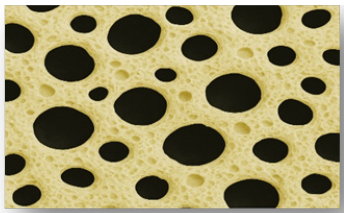


Fig 13.
Detailed microscopic image of bone cement porosity.

Causes of Increased Porosity:

1.Air Trapping: This issue is particularly exacerbated during manual handling of bone cement.

2.Improper Mixing Techniques: Failure to employ advanced methods to minimize air bubble formation.

Factors Reducing Porosity:

1.Centrifugation: Utilizing centrifugal force to reduce air bubbles.

2.Vacuum Mixing: Lowering the ambient pressure to prevent the formation of air bubbles.

3.Pressurization: Applying pressure to compress the cement and reduce void spaces.

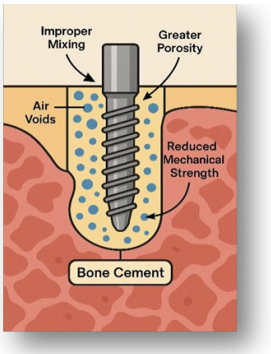


Fig 14.
The presence of air bubbles in bone cement caused by improper mixing leads to increased porosity and decreased mechanical properties and durability of the implant.

Hysteresis in Viscoelastic Materials

Biomaterials are subjected to various mechanical loads and stresses during physiological activity and therefore require an appropriate degree of fluidity and deformability. These properties enable them to withstand applied stresses and maintain optimal functionality.

Viscoelastic Behavior

Viscoelastic materials, such as bone, exhibit a combination of viscous (adhesive) and elastic (spring-like) behavior. Upon loading, these materials initially undergo elastic deformation, followed by plastic deformation. During this process, not all of the input energy is recovered; part of it is absorbed by the material. This phenomenon is known as hysteresis.

Hysteresis

Hysteresis refers to the energy loss during loading and unloading cycles. In viscoelastic materials, when a force is applied, the material deforms, and after the removal of the force, it gradually returns to its original shape. However, a portion of the energy is dissipated as heat during this process.

Comparison of Elastic and Viscoelastic Material Behavior

1. Elastic Materials (e.g., stone, glass, ceramics):

- These materials obey Young's modulus and the laws of elasticity.
- Deformation occurs within the material's elastic limit and is fully reversible upon removal of the force.

- When stress exceeds the material's threshold, it fails in a brittle and sudden manner.

2. Viscoelastic Materials (e.g., bone and certain polymers):

- These materials deform under applied forces and gradually return to their original shape once the force is removed.
- A portion of the mechanical energy is lost as heat during each load-unload cycle due to internal friction.

Fatigue in Materials

Fatigue strength is a measure of a materials or structure's ability to withstand fatigue failure, which occurs when an object or prosthesis is subjected to cyclic stresses or repeated loading over time. Fatigue failure can take place even when the applied stresses are below the material's ultimate tensile strength.

Fatigue Failure

Fatigue failure typically occurs in materials and structures exposed to fluctuating loads or vibrations. This type of failure often happens suddenly and without prior warning, as fatigue-induced damage accumulates gradually at the microscopic level.

Stages of Fatigue Failure

1.Crack Initiation:

Microcracks form on the surface of the material, usually at stress concentration points.

2.Crack Propagation:

These cracks progressively grow with each loading cycle.

3.Final Fracture:

When the crack reaches a critical size, the material fails catastrophically.

Factors Affecting Fatigue

- **Stress Amplitude:**

Higher stress amplitudes reduce the fatigue life of the material.

- **Number of Loading Cycles:**

Increasing the number of cycles raises the likelihood of fatigue failure.

- **Material Properties:**

Materials with high strength and toughness generally exhibit better fa-

tigue resistance.

- **Environmental Conditions:**

Factors such as temperature, humidity, and corrosion can significantly influence fatigue performance.

Applications and Importance

- In the design of medical prostheses and implants, fatigue resistance is crucial, as these components are subject to repetitive loading from daily activities.
- In aerospace, automotive, and civil engineering industries, evaluating a material's fatigue strength is essential to ensure the safety and durability of structures.

Summary of Orthopedic Cements (Chapters 1–7)

In general, four main types of bone cements are used in orthopedic applications:

1. **Acrylic-based cement (PMMA – Polymethyl Methacrylate)**
2. **Polypropylene fumarate-based cement (PPF)**
3. **Calcium phosphate cement (CPC)**
4. **Glass ionomer cement (GIC)**

The first two types (PMMA and PPF) are polymer-based, while the latter two (CPC and GIC) are ceramic-based. Among them, PMMA is the most widely used cement in both the medical and dental fields.

PMMA (Polymethyl Methacrylate)

History:

PMMA is a transparent thermoplastic polymer first developed in 1928 by Rohm & Haas. It has numerous modern applications, including:

- Optical lenses
- Acrylic nails
- Paint and coatings industry
- Security barriers
- Medical devices
- LCD screens
- Everyday consumer products

Applications in Orthopedics:

Due to its easy accessibility, cost-effectiveness, and wide availability, PMMA is commonly used in:

- **Arthroplasty** (joint replacement)
- **Cranioplasty** (cranial reconstruction)
- **Kyphoplasty** (treatment of vertebral fractures)

Major Applications of Orthopedic Cements

Orthopedic bone cements are primarily used in four categories:

1.Arthroplasty:

- In joint replacement surgeries (especially hip and knee), for both uni-compartmental and total arthroplasties.

2.Tumors:

- In the treatment of bone tumors, particularly metastatic lesions, cysts,

and tumors like giant cell tumor (GCT).

3.Kyphoplasty:

- For managing vertebral compression fractures due to osteoporosis.

4.Fractures:

- Applied in pathologic and osteoporotic fractures.

Special Applications of Bone Cements

1.Rheumatoid Arthritis (RA)
2.Avascular Necrosis (AVN)
3.Sickle Cell Anemia (due to joint destruction)
4.Calcaneal Fractures and Calcaneal Tumors
5.Bone Tumors
6.Infected Joints (Two-stage revision arthroplasty)
7.Chronic Osteomyelitis
8.Open (acute) fractures
9.Hip and Knee Arthroplasty

This summary outlines the types, compositions, and clinical uses of orthopedic bone cements, with an emphasis on PMMA, the most prevalent formulation in current practice.

Historical Milestones

- **1870 – Themistocles Gluck:**
  He was the first surgeon to use ivory and natural materials like pine resin for knee joint replacement.
- **Early 1900s – Otto Rohn and Kulzer:**
  Chemists in Germany and England conducted studies on the physical and chemical properties of orthopedic cements. Their work led to advancements in the polymerization process and cement curing.
- **1960s:**
  Orthopedic cements transitioned from dental and cranial applications into the field of arthroplasty.
- **2003:**
  Bone cement was approved for commercial use in the United States and Europe.
- **2004:**
  Gentamicin-loaded bone cement received approval for use in two-stage revision surgeries to manage joint infections.

- **2006:**
  The second-, third-, and fourth-generation orthopedic cements were developed, incorporating improvements in performance and biocompatibility.

Recent Advancements

- **Incorporation of Antibiotics:**
  Gentamicin was the first antibiotic added to bone cement to reduce the risk of postoperative infections.
- **Improved Mixing Technologies:**
  Innovations in cement mixing tools and viscosity control have led to the production of more diverse and higher-quality products, enhancing handling properties and clinical outcomes.

These developments highlight the evolution of bone cements from basic natural materials to highly engineered, antibiotic-enhanced biomaterials, playing a pivotal role in modern orthopedic surgery.

Chapter 8

Review of Scientific Literature on Orthopedic Bone Cement

Chapter 8: Review of Scientific Literature on Orthopedic Bone Cement

1.Effects of Normal Saline on the Biomechanical Properties of Bone Cement

- Studies indicate that the use of normal saline (isotonic saline solution) during the mixing or irrigation stages of orthopedic cement can influence its mechanical properties.
  - **Reduction in compressive strength:** Some studies have reported that exposure of bone cement to normal saline may cause a temporary reduction in compressive strength, particularly during the early phases of setting.
  - **Viscosity changes:** Saline may reduce the viscosity of the cement and facilitate its application process; however, this decrease might result in increased porosity.
  - **Recommendation:** Limited use of saline is advised when initial cement strength is critical (e.g., load-bearing arthroplasty), and advanced techniques such as vacuum mixing are suggested to minimize porosity.

2.Replacement of Orthopedic Cement with Chitosan Polymer

- Chitosan, a natural polymer derived from chitin, has been studied as a biocompatible alternative to traditional cements such as PMMA (Polymethyl methacrylate).
  - **Advantages:**
    - Biodegradability and reduced long-term complications.
    - Intrinsic antimicrobial properties of chitosan that reduce infection risk.
    - Capability to stimulate bone regeneration (osteoinductive property).
  - **Challenges:**
    - Lower mechanical strength compared to PMMA, limiting its use in high-load applications.
    - Requirement for chemical modification to improve adhesion to bone tissue.
  - **Suggestion:** Combining chitosan with nanoparticles (e.g., hydroxyapatite) or synthetic polymers to enhance mechanical properties.

3.Antibiotic Release from Orthopedic Cement

- Antibiotic-loaded bone cements (e.g., containing gentamicin or vancomy-

cin) are used for the prevention or treatment of postoperative infections.

- **Factors affecting release:**
  - Cement porosity: Higher porosity increases antibiotic release.
  - Type of antibiotic: Smaller molecules (like gentamicin) are released more rapidly.
  - Mixing method: Vacuum or centrifuge mixing can improve the uniform distribution of antibiotics.
- **Key findings:**
  - Initial burst release during the first 72 hours, followed by a gradual decline over several weeks.
  - Subtherapeutic surface concentrations of antibiotics after a few weeks may lead to bacterial resistance (low-dose antibiotic exposure).
- **Strategy:** Use of antibiotics with longer half-lives or multi-drug loading formulations.

4.Optimization of Orthopedic Cement Use

- **Optimization of application techniques:**
  - **Pressurization of cement:** Controlled pressure to allow cement penetration into trabecular bone and reduce porosity.
  - **Advanced mixing:** Use of centrifuge or vacuum devices to reduce air bubbles.
  - **Control of polymerization temperature:** Lowering reaction temperature to prevent thermal necrosis of surrounding tissues.
- **Development of next-generation cements:**
  - Smart cements with controlled drug release capabilities.
  - Bioactive cements that bond with bone tissue.

5.Essential Recommendations for Clinical Use

- **Selection of appropriate cement:**
  - Use antibiotic-loaded cements in high-infection-risk procedures (e.g., reconstructive surgeries).
  - Employ low-exotherm cements in temperature-sensitive surgeries.
- **Improvement of mixing and application techniques:**
  - Avoid manual mixing when porosity is a critical factor.
  - Utilize pressurization to enhance cement-bone adhesion.
- **Clinical monitoring:**
  - Regular assessment of antibiotic release and blood concentrations in long-term use.

- Monitor for signs of early failure (e.g., prosthesis loosening) in high-load patients.

- **Future research:**
  - Development of hybrid cements combining natural and synthetic polymers.
  - Long-term studies on the biocompatibility of new-generation cements.

This review demonstrates that recent advancements in orthopedic cement, including the use of novel materials and application techniques, can help overcome clinical challenges and improve surgical outcomes.

"Effect of Saline Solution Contamination on the Mechanical Properties of Bone Cement with Medium and High Viscosity."

Effect of physicochemical properties of saline solution contamination on selected mechanical properties of seasoned Acrylic Bone Cement of medium-high viscosity [material - MDPI]

Recent studies indicate that contamination of bone cement with saline solution (physiological saline) can have significant effects on its mechanical properties. This issue is particularly important for cements with medium and high viscosity, such as CMW1 and CMW3. The following section reviews these effects and the findings of related studies:

1. Factors Affecting the Durability of Bone Cement

- **Proper Prosthesis Fixation:**
  Precise fixation of the prosthesis with bone cement inside the bone canal and using cement with appropriate mechanical properties are key to successful orthopedic surgeries.
- **Mechanical Loads and Aggressive Environment:**
  After implant placement, the cement is subjected to mechanical loads caused by patient movement and a biologically aggressive environment.
- **Common Surgical Contaminations:**
  Normal saline (physiological saline) and Ringer's lactate solution are commonly used for washing the bone canal and cooling the cement immediately after placement.
  These solutions may penetrate the cement and alter its mechanical properties.

2. Effects of Saline Solution Contamination on Mechanical Properties of Cement

a) Reduction in Microhardness

- **Observations:**
  Increasing contamination with saline solution leads to a significant decrease in the microhardness of the cement.Contaminated cements exhibit up to a 10% average reduction in mechanical strength.
- **Seasoning Time:**
  The decrease in mechanical strength depends on the seasoning time (duration the cement remains in solution).
  CMW1 cement degrades faster than CMW3.

b) Reduction in Compressive Strength

- **Statistical Results:**
  CMW1 cement samples showed a significant reduction in compressive strength compared to CMW3.
  The compressive strength reduction was especially notable after 15 days of exposure to saline solution.
- **Effect of Contamination Percentage:**
  Samples with contamination exceeding 6% showed greater reductions in microhardness and compressive strength.

c) Surface Degradation

- **Impact on Prosthesis Longevity:**
  Surface degradation of the cement may lead to implant loosening and reduced prosthesis lifespan. The formation of free space between cement and bone may provide a suitable environment for bacterial colony formation.

3. Comparison of CMW1 and CMW3 Cements

Property	CMW1	CMW3
Viscosity	Medium	High
Resistance to Contamination	Lower	Higher
Degradation Rate	Faster	Slower
Compressive Strength	Significant Decrease	Moderate Decrease
Micro hardness	Greater Reduction	Lesser Reduction

4. Mechanisms of Cement Degradation

1.Penetration of Saline Solution:

Saline solution infiltrates the cement, causing swelling and decreased density, leading to reduced mechanical strength and microhardness.

2.Surface Degradation:

Saline penetration induces microcracks and surface degradation that can propagate over time, resulting in prosthesis loosening.

3.Formation of Free Space:

Decreased adhesion between cement and bone creates free spaces that can serve as sites for bacterial colonization and infection.

5. Conclusion and Recommendations

- **Reduction in Mechanical Properties:**
  Saline contamination causes decreased microhardness and compressive strength, especially in medium-viscosity cement (CMW1).
- **Surface Degradation and Prosthesis Loosening:**
  Surface degradation may cause prosthesis loosening and reduced implant longevity.
  Free space formation increases the risk of infection.
- **Recommendations:**
  - Minimize cement exposure to saline solution during surgery.
  - Use cements more resistant to contamination (e.g., CMW3).
  - Closely monitor patients for early signs of prosthesis loosening or infection.

Key References for Further Study

- MDPI Materials: Study on the effect of saline solution contamination on the mechanical properties of bone cement.
- Comparative analysis of contamination resistance and degradation between CMW1 and CMW3 cements.

Acrylic Bone Cement Incorporated with Low-Molecular-Weight Chitosan

[Polymers – MDPI]

The use of chitosan as an additive in acrylic bone cement (PMMA) has attracted attention due to its favorable biological and mechanical properties. Chitosan is a natural polymer derived from the deacetylation of chitin, exhibiting

characteristics such as biocompatibility, bioactivity, and biodegradability. The following reviews the effects of adding low molecular weight chitosan to bone cement and its influence on mechanical and biological properties:

1. Effects of Chitosan on Bone Cement Properties

a) Increase in Viscosity and Porosity

- **Chitosan Structure:**
  Chitosan possesses a semi-crystalline structure with glucosamine units that increase the viscosity of the cement. This increased viscosity leads to the formation of voids during polymerization where air is trapped, resulting in higher cement porosity.
- **Impact on Mechanical Properties:**
  Increased porosity causes a reduction in mechanical properties such as compressive strength and microhardness.
- **Impact on Bioactivity:**
  The increased porosity and surface roughness enhance cell adhesion and bone tissue growth, thereby improving the cement's bioactivity.

b) Reduction in Polymerization Temperature (Tmax)

- **Mechanism:**
  Due to intramolecular hydrogen bonding, chitosan acts as a heat sink, absorbing the heat generated during polymerization.
  This results in a lowered maximum polymerization temperature (Tmax).
- **Benefits:**
  Reduced polymerization temperature decreases the risk of thermal necrosis to surrounding tissues, which is particularly important in orthopedic surgeries requiring close contact between cement and living tissue.

c) Increase in Setting Time (Tset)

- **Mechanism:**
  Adding chitosan prolongs the setting time of the cement.
- **Benefits:**
  This extension provides surgeons with more time for shaping and accurate placement of the cement, enhancing operative efficiency and reducing errors due to limited working time.

2. Effects of Chitosan on Biocompatibility

a) Hydrophilicity

- **Water Contact Angle Measurement:**
  The hydrophilicity of cement surfaces is assessed by measuring the water contact angle.
  A lower contact angle indicates higher hydrophilicity and improved biocompatibility.
- **Effect of Chitosan Molecular Weight:**
  For chitosan loading less than 10%, the water contact angle increases with chitosan content.
  For loadings greater than 10%, the water contact angle decreases due to the increased number of hydrophilic groups.
- **Improvement in Biocompatibility:**
  Reduced water contact angles in cements with 15% and 20% molecular weight chitosan enhance biocompatibility.

b) Enhanced Cell Adhesion and Bone Tissue Growth

- Increased porosity and surface roughness improve cellular adhesion and promote bone tissue growth.
- This contributes to forming a strong mechanical bond between cement and bone.
- Bone ingrowth within the cement improves implant stability and reduces the risk of loosening.

3. Summary and Conclusions

- Adding chitosan to bone cement:
  - Increases viscosity and porosity.
  - Decreases mechanical properties such as compressive strength, but improves bioactivity and biocompatibility.
- Reduction of polymerization temperature (Tmax) lowers the risk of thermal necrosis.
- Increased setting time (Tset) provides surgeons more working time with the cement.
- Improved biocompatibility through lower water contact angles and enhanced hydrophilicity promotes better cell adhesion and bone tissue growth.

Recommendations

Use chitosan with appropriate molecular weight (15% to 20%) to optimize biocompatibility and reduce polymerization temperature.

Control cement porosity to maintain a balance between mechanical and biological properties.

Closely monitor patients to ensure implant stability and prevent loosening.

Clinical Applications and Final Remarks

Key Findings:

1.Calcium (Ca) and Phosphorus (P) Ion Storage:

Formulations containing more than 15% molecular weight chitosan show calcium ion storage from the first week and phosphorus ion storage from the fourth week after immersion in simulated biological fluid. In contrast, formulations with less than 15% chitosan show ion storage between the second and twelfth weeks.

2.Faster Bioactivity:

To achieve faster bioactivity in biological environments, the minimum chitosan loading should be 15% molecular weight.

3.Improved Physical and Mechanical Properties:

Formulations with 15% or higher chitosan loading demonstrate:

- Greater thermal stability (reduced maximum polymerization temperature).
- Increased porosity.
- Higher surface roughness.
- Better hydrophilicity (lower water contact angle).
- Longer setting time (increased Tset).

Clinical Uses:

Based on these findings, chitosan-incorporated cements with 15% or higher molecular weight are suitable for:

- **Bone Fillers:** For low-load-bearing bone defect filling.
- **Vertebroplasty:** Stabilization of fractured or weakened vertebrae.
- **Balloon Kyphoplasty:** Restoration of compressed vertebral height and pain reduction due to spinal fractures.

Final Summary:

- Chitosan-incorporated acrylic bone cements with 15% or higher molecu-

lar weight present an ideal option for applications requiring rapid bioactivity, high porosity, and low load-bearing capacity.

- These cements have promising applications, especially in spinal surgeries and bone filler materials.

Controlling Antibiotic Release from Polymethylmethacrylate (PMMA) Bone Cement

[Biomedicines – MDPI]

Clinical Problem: Periprosthetic Joint Infection (PJI)

Prosthetic Joint Infection (PJI) can lead to significant mortality, with its severity estimated to be comparable to that of Ewing Sarcoma. Common complications of PJI include the risk of amputation, excessive bleeding, prolonged hospitalization, and the need for multiple surgical interventions.

- **Prevalence and Significance:**
  Periprosthetic Joint Infection (PJI) is one of the most common causes of failure in total knee arthroplasty and the third leading cause in hip revision surgeries. The reported incidence of PJI ranges from 0.2% to 2%.
- **Infection Mechanism:**
  PJI results from bacterial biofilm formation on biomaterial surfaces (such as implants or cement) or surrounding tissues. This biofilm protects bacteria from the host immune system and antibiotic treatment, facilitating bacterial proliferation and persistence.

Solution: Antibiotic-Loaded Bone Cement

- **History and Application:**
  The use of antibiotic-loaded bone cements, especially PMMA-based, has long been recognized as an effective approach to reduce post-surgical infection risk or to treat existing infections. These cements also lower the rate of infection recurrence.
- **Current Challenges:**
  Existing antibiotic-loaded cements require improvements in the following areas:

1.Antibiotic Selection: Choosing antibiotics with an appropriate spectrum of activity and longer half-life.

2.Mixing Techniques: Enhancing mixing methods to achieve uniform antibiotic distribution within the cement.

3.Release Profile: Controlling the amount and duration of antibiotic release to prevent tissue toxicity and bacterial resistance.

Factors Affecting Antibiotic Release

1. Effective Elution Volume:

- The volume of fluid available for antibiotic elution from the cement is one of the most important predictors of antibiotic concentration in the in vivo environment.
- Although difficult to calculate precisely, anatomical data estimate the effective elution volume to be approximately 170 mL for the hip joint and 90 mL for the knee joint.

2. Fluid Absorption and Reabsorption:

- The amount of fluid absorbed and subsequently reabsorbed in clinical environments remains unknown.
- Accurate calculation of these values requires clinical data and detailed reports of local and systemic antibiotic concentrations over time.

Recommendations for Future Research

1. Development of Optimized Formulations:

- Design antibiotic-loaded bone cements with controlled and sustained release profiles.
- Utilize antibiotics with longer half-lives to reduce dosing frequency.

2. Improvement of Mixing and Cleansing Methods:

- Develop advanced mixing techniques to achieve uniform antibiotic distribution within the cement.
- Investigate the effects of cleansing methods on tissue toxicity and mechanical strength of the cement.

3. In Vitro and In Vivo Evaluation:

- Conduct in vitro experiments to simulate physiological conditions.
- Perform clinical (in vivo) studies to assess the efficacy and safety of new cement formulations.

4. Measurement of Antibiotic Concentrations:

- Simultaneous measurement of local and systemic antibiotic concentrations over time to better predict release profiles.

Final Summary

- Antibiotic-loaded bone cements play a vital role in the prevention and treatment of periprosthetic joint infections.
- To enhance their performance, targeted research is needed on formulation optimization, mixing techniques, and clinical evaluation.
- Precise control of antibiotic release and reduction of tissue toxicity are key to achieving better clinical outcomes and prolonging implant longevity.

Alternative Formulations of Acrylic Bone Cement for Cemented Arthroplasty: Current Status, Predictions, and Future Expectations

Alternative Acrylic Bone Cement Formulation for Cemented Arthroplasty Present Station, Key Issues & Future Prospect [Research Gate]

Current Status

Acrylic bone cements based on polymethylmethacrylate (PMMA), despite their widespread use in cemented arthroplasty, face several challenges:

1.High polymerization temperature reaching up to 120°C, which can cause thermal necrosis of surrounding tissues.

2.Reduced bioactivity, characterized by a lack of ability to stimulate bone regeneration.

3.Volumetric shrinkage during curing, leading to increased porosity and decreased adhesion to bone.

4.Cellular toxicity caused by released monomers.

Chapter 9

New Generations of Bone Cement

Chapter 9: New Generations of Bone Cement

Sixteen Alternative Formulation Groups for Enhancing PMMA Bone Cements

Recent research has led to the development of alternative formulations aimed at addressing the limitations of conventional PMMA bone cements. These innovations can be categorized into sixteen major groups:

1. Low-DMPT Content Cement

- **Advantages:**
  - Increases setting time (t_{set}) by up to 54%.
  - Reduces maximum polymerization temperature (T_{max}) by up to 7%.
  - Decreases residual monomers and improves in vivo stability.
- **Disadvantages:**
  - Up to 4% reduction in compressive strength (UCS).
  - Polymerization rate at 37°C drops by up to 97%.

2. Two-Solution Cement

- **Advantages:**
  - Pore-free structure.
  - Enhanced compressive strength and flexural modulus.
  - Reduced in vivo residual monomer content.
- **Disadvantages:**
  - Higher monomer volume may reduce handling efficiency.

3. Antioxidant Cement (Vitamin E-Modified)

- **Advantages:**
  - Decreased T_{max}.
  - Extended t_{set}.
- **Disadvantages:**
  - Increased residual monomers.
  - Reduced compressive strength (UCS).

4. Cement with Alternative Accelerators

- **Advantages:**
  - Replaces DMPT with alternatives such as DMAL or DMAO.
  - Enhances thermal and mechanical properties.

5. Cement with Modified Monomers

- **Advantages:**
  - Shorter setting time (t_{set}).
- **Disadvantages:**

- Elevated residual monomers.
- Reduced compressive strength (UCS).

6. Cement with Uniform Radiopaque Distribution

- **Advantages:**
  - Incorporation of coated or uncoated barium sulfate.
  - Increased compressive and tensile moduli and improved fatigue life.

7. Cement with Alternative Radiopacifiers

- **Advantages:**
  - Improved thermal and mechanical performance.
- **Disadvantages:**
  - Some unwanted or negligible effects.

8. Crosslinked Cement

- **Advantages:**
  - Formation of an insoluble network, enhancing stiffness and toughness (K_IC).
  - Reduces volumetric shrinkage.
- **Disadvantages:**
  - Dependent on the type of crosslinking agent (e.g., PEG-DMA or PEG-DMAA).

9. Solid-Phase MMA/HPMA Cement

- **Advantages:**
  - Shortens setting time (t_{set}).
- **Disadvantages:**

  Increases T_{max}.

10. Cement with Enhanced Adhesion

- **Advantages:**
  - Reduces interface gaps between bone/implant and cement due to shrinkage.

11. Low-Modulus Cement

- **Advantages:**
  - Decreases interfacial stress between the implant and bone.
  - Reduces bone resorption and fibrous membrane formation.
  - Lowers the risk of aseptic loosening.

12. Reinforced Cement

- **Advantages:**
  - Improves key mechanical properties such as fatigue life, toughness (K_IC), tensile strength (UTS), and compressive strength (UCS).

13. Partially Degradable Cement

- **Advantages:**

- Promotes bone ingrowth into pores resulting from cement degradation.

14. Biodegradable Cement

- **Advantages:**
  - Lower T_{max}.
  - Incorporates inert biodegradable fillers.

15. Bioactive Cement

- **Advantages:**
  - Improves characteristics such as T_{max} and fatigue life.
- **Disadvantages:**
  - Increased water uptake.

16. Mixed-Functionality Cement

- **Advantages:**
  - Combines multiple properties based on the types of added particles.

Predictions, Future Expectations, Ongoing Studies, and Emerging Products

1.Gradual Replacement of Conventional PMMA:

- Transition toward bioactive and smart cements that offer both mechanical strength and biological activity.

2.Personalized Therapy:

- Use of 3D printing and nanotechnology to fabricate implants tailored to patient-specific anatomy.

3.Controlled Drug Release:

- Development of cements with stepwise release profiles or light-activated antimicrobial functions.

4.Reduced Side Effects:

- Non-toxic, low-exothermic formulations to avoid thermal damage to surrounding tissues.

5.Integration with Tissue Engineering:

- Combining cements with nanofiber scaffolds or growth factors to facilitate bone regeneration.

Conclusion

Alternative formulations of acrylic bone cement represent the most promising strategies to overcome the limitations of traditional PMMA-based materials. Shortly, smart, bioactive, and patient-specific cements are expected to become the new standard in orthopedic surgery.

Chapter 10

Advances in Orthopedic Bone Cement and the World of Biomaterials

Chapter 10: Advances in Orthopedic Bone Cement and the World of Biomaterials

Learning point in the Development of B.C

1. Introduction

Orthopedic bone cement based on polymethyl methacrylate (PMMA) serves as a fundamental biomaterial in joint replacement surgeries and bone reconstruction, playing a key role in ensuring mechanical stability and facilitating prosthesis–bone integration.

2. Historical Development

- **1928:** Hass and Rohn first reported the synthesis of PMMA.
- **1935:** The Kulzer company introduced a formulation combining MMA monomer and PMMA powder for mass production.
- **1938:** Animal trials on cranial monkey models demonstrated the capability of the initial formulation to repair cranial defects.
- **1943:** The PMMA polymerization reaction was accepted as a standard method for bone fixation in orthopedic surgeries.
- **1953:** Chanley demonstrated the efficacy of PMMA cement in hip and knee arthroplasty.

3. Formulation and Polymerization Mechanism

Orthopedic cement consists of a liquid monomer (Methyl Methacrylate) and a PMMA powder, along with a radical initiator. The polymerization reaction is carried out under cold-curing conditions.

The heat generated during polymerization can reach 100°C in the lab and up to 65°C in biological media (blood, serum, tissue), which, if not thermally managed, may lead to thermal necrosis. (Saline is used to cool the cement during the hardening phase.)

4. Physical Bonding Mechanism

The spongy structure of bone allows the polymeric cement to penetrate microscopic pores, thereby providing mechanical fixation. Thus, PMMA cement acts as a grout, not a glue, and does not chemically bond with the bone matrix.

5. Clinical Applications

- **Hip and Knee Arthroplasty:** PMMA is used to ensure initial prosthesis stability and reduce micromotion at the joint.
- **Cranioplasty and Cranial Defect Reconstruction:** Provides a biomechanical filler with proper defect adaptation.
- **Complex Fracture Repair and Bone Grafting:** Offers a scaffold for secondary bone regeneration.

6. Thermal Effects and Necrosis

The exothermic reaction of polymerization may lead to thermal necrosis if temperatures exceed 60°C. Thus, the use of thermal modifiers and cement cooling protocols during implantation is essential.

7. Biocompatibility Criteria and Clinical Regulations

- Free from systemic and local toxicity
- Does not elicit chronic immune or allergic responses
- Non-cytotoxic and non-carcinogenic
- Requires clinical efficacy validated by controlled clinical trials
- The surgical team must be trained and competent in handling the physical, chemical, and safety aspects of monomer exposure

8. Complications and Clinical Challenges

- **Bone Cement Implantation Syndrome (BCIS):** • Associated with hemodynamic fluctuations and pulmonary complications
- **Cement Disease:** Due to chronic inflammatory responses around the material
- **Prosthesis Loosening:** Resulting from microstructural bonding failure and mechanical wear
- **Environmental Hazards:** Exposure to monomer vapors requires efficient fume extraction systems

9. Research Outlook

Current research focuses on:

- Incorporation of nanomaterials
- Use of local anti-inflammatory agents
- Osteoinductive accelerators to reduce polymerization temperature, enhance flexural and compressive strength, and improve biological response

10. Versatile Applications of Biomaterials in Modern Medicine

Due to degenerative diseases and tissue wear, biomaterials now play a critical role in replacing and regenerating dysfunctional tissues. PMMA is widely applied clinically due to its mechanical strength and acceptable biocompatibility:

- **Dental Tools:** Used as a filling material and for temporary prostheses
- **Skin Repairs (Bandages):** Acts as a volume-maintaining filler in modern dressings
- **PMMA Medical Catheters:** Due to dimensional stability and chemical resistance, PMMA is used in manufacturing the rigid components of catheters.
- **Orthopedic Plates & Implants:** PMMA-based compounds are employed in certain orthopedic plates and implants that require high stiff-

ness and strength.

- **Coating for Contact Lenses & Arthroplasty:** Thin PMMA coatings can reduce friction surfaces and prevent protein deposition on lenses or prostheses.
- **Cardiovascular Implants (Valves & Stents):** PMMA is used in some non-movable parts of cardiac implants to create a rigid structure and maintain precise sizing.
- **Advanced Wound Dressings:** Combining PMMA with hydrogels provides a controlled moist environment conducive to wound healing.
- **Connection Devices:** PMMA serves as a stable and biocompatible bonding material in devices for fluid transfer or optical fibers inside the body.

11. Standard Types of Bone Cements

Currently, various bone cements have been developed based on clinical needs and biomechanical properties:

- **PMMA (Poly Methyl Methacrylate):** Traditional cement with high compressive strength and adjustable working time, widely used in hip and knee arthroplasty.
- **Calcium Sulfate Cements (CSCs):** These cements are used for their controlled solubility and facilitation of bone resorption in fracture repair and bone grafting.
- **Calcium Phosphate Ceramic Cements:**
  - **Hydroxyapatite (HA):** Recognized as the primary scaffold in bone tissue engineering, it possesses a structure similar to the mineral matrix of natural bone.
  - **Tricalcium Phosphate (TCP):** Used in bone regeneration due to its appropriate resorption rate and ability to stimulate osteoclast activity.
- **Organic–Inorganic Composite Cements:** By combining polymeric materials with ceramic phases, these cements optimize the balance between flexibility and mechanical strength, enabling controlled release rates of bone growth factors.

Comprehensive Resources for Further Study:

1.Chanley J. “PMMA Bone Cement in Hip Arthroplasty,” Journal of Bone and Joint Surgery, 1953.
2.Lewis G. “Properties of Acrylic Bone Cement,” Orthopedic Biomaterials Review, 2008.
3.Khandelwal S., Kanupur A. “Advances in Orthopedic Biomaterials,” International Journal of Biomaterials, 2021.

Viscosity of Orthopedic Cement

In the studies by Dunne, it has been shown that simultaneously with the onset of the setting process of orthopedic cement—and in some formulations even during the pre-setting phase—viscoelastic changes occur in the material. The viscosity of the cement affects two key aspects of clinical performance:

1.Handling Capability

- **Low Viscosity:**
  - Low viscosity increases the waiting time, meaning that after the mixing is completed, there is a longer opportunity for injection and placement before initial hardening begins. However, the working time is short, and the paste quickly separates from the glove or syringe surfaces.
- **High Viscosity:**
  - High viscosity reaches the desired consistency immediately after mixing and does not require a waiting time; this facilitates ease of use and reduces the risk of displacement during placement.

2.Penetration and Fixation Quality

- **Low Viscosity:**
  - Low viscosity enables deeper penetration into the pores and microstructures of cancellous bone and prevents the backflow of blood and tissue fluids into the injection space. This feature plays a vital role in improving micro-interlocking and achieving physical (non-chemical) fixation of the prosthesis.
- **High Viscosity:**
  - Although high viscosity reduces local penetration, it allows better control of cement flow and minimizes the risk of unwanted extrusion into surrounding tissues.

Effect on Longevity and Loosening

- The use of cement with optimal viscosity ensures maximum mechanical engagement at the prosthesis–bone interface and reduces subsequent micromovements, leading to improved long-term fixation and a lower rate of loosening.
- The appropriate viscosity range should be selected based on the type of prosthesis, the volume of the void, and the injection technique (syringe or pressure injector) to achieve the best combination of penetration and handling.

Conclusion:

In clinical practice, determining the ideal viscosity of orthopedic cement should be based on the specific needs of the procedure—including the type of joint, the placement method, and the condition of the bone tissue—in order to ensure both ease of use and durability of fixation.

Antibiotics Used in Antibiotic-Impregnated Bone Cement (AIBC):

Antibiotics used in antibiotic-impregnated bone cement must possess the following characteristics to ensure prophylactic or therapeutic efficacy against infection without compromising the mechanical properties of the cement:

1.Broad Spectrum:

The selected antibiotic should be effective against common Gram-positive and Gram-negative microorganisms involved in orthopedic infections and, if possible, have the ability to inhibit the growth of resistant bacteria.

2.Pure Powder Form:

The antibiotic should be supplied as a highly pure powder with fine particles (≤ 50 microns) to enable uniform distribution within the cement matrix during mixing.

3.Optimal Drug Elution:

The controlled release property must allow a high and effective drug dose within the first 24–48 hours post-injection, followed by a stable and prolonged release lasting several weeks.

4.Minimal Allergic Reaction:

Both systemic and local safety are essential; the antibiotic must not induce skin hypersensitivity or local inflammatory reactions.

5.Preservation of Mechanical Properties:

The added antibiotic should not cause a significant reduction in compressive strength (> 70 MPa), flexural strength, or physical adhesion of the cement.

Common Antibiotics Used for Orthopedic Cement Impregnation:

- **Gentamicin Sulfate:**

Broad-spectrum activity against Staphylococci and Gram-negative bacteria, suitable drug delivery, release profile, and minimal impact on cement strength.

- **Vancomycin HCl:**

Effective against resistant Gram-positive microorganisms (including MRSA), supplied as pure powder with slow-release characteristics.

- **Tobramycin Sulfate:**

Similar spectrum to gentamicin, sometimes combined with uridine to improve the release profile.

In clinical practice, depending on the microbial sensitivity and regional resistance patterns, doses of 1–2 grams of gentamicin per 40-gram PMMA cement package or up to 4 grams of vancomycin in suspected MRSA cases can be used; however, the final dosage should be adjusted according to the cement manufacturer's guidelines and patient safety considerations.

Chapter 11

Recent Advances in the Design and Formulation of Bone Cements

Chapter 11: Recent Advances in the Design and Formulation of Bone Cements

Recent advances in materials science and biomaterials have led to the development of bone cements with enhanced performance, improved safety, and superior physical and chemical properties. These innovations aim to reduce side effects, increase tissue compatibility, and improve mechanical conformity with bone.

1.Use of novel monomers such as DMPT

Newer monomers like DMPT have reduced setting time by 54% and decreased peak polymerization temperature by 7%. However, their polymerization rate at 37°C has dropped by 97%, which requires further investigation.

2.Two-solution cements

Cements formulated from two solutions (e.g., base and activator) increase flexural strength, reduce porosity, and improve uniformity.

3.Addition of Vitamin E

Formulations containing Vitamin E provide antioxidant properties and reduce the peak exothermic temperature during polymerization, effectively decreasing thermal necrosis.

4.Replacement of DMPT with DMAL or DMAO

Using alternative monomers such as DMAL or DMAO instead of DMPT reduces polymerization acceleration rates, enabling better control over the curing process.

5.Modified monomer cements

Modified monomers enhance mechanical properties, including fracture resistance and adhesion to implants.

6.Cement with uniform radiopaque dispersion

Employing uniformly dispersed radiopaque particles (e.g., barium sulfate or zirconia) improves imaging contrast without compromising structural strength.

7.Cross-linked cements

The presence of cross-linkers stabilizes the polymer network and increases durability against hydrolytic degradation.

8.Solid phase cements

Formulations incorporating a solid phase increase stability and homogeneity while reducing heat release rate.

9.Surface adhesion modification

Controlled adhesion to tissue or implants by surface modification of cement particles decreases loosening and inflammation.

10.Reduced elastic modulus

Designing cements with a lower elastic modulus enhances mechanical compatibility with host bone and reduces fracture risk.

11.Reinforced cements

Additives such as fibers or nanoparticles (e.g., Nano silica or nanohydroxyapatite) are used to improve mechanical strength.

12.Partially degradable cements

Gradual degradation capability is useful in scenarios requiring progressive replacement by new bone.

13.Biodegradable cements

Fully biodegradable cement designs, especially for temporary or non-load-bearing applications, represent an emerging research avenue.

14.Bioactive cements

These cements exhibit high biological activity, stimulating bone tissue regeneration and accelerating osseointegration.

15.Hybrid-function cements

Multi-functional formulations (e.g., drug delivery, mechanical and biological compatibility simultaneously) are considered the next generation of bone cements.

References:

Biomedicine – Application of DMPT in bone cement formulations
ResearchGate – Collection of articles on novel monomers and additives in biomaterial cements

Body reaction to orthopedic cement and Bone Cement Implantation Syndrome (BCIS)

1.Prevalence and Risk Factors

- Increased risk in cardiovascular patients and those classified as ASA III and IV.
- A cohort study (2006–2007) with a mean follow-up of 918 days reported an average hospital stay of 8.6 days for the cemented group; mortality was 2.3% in cemented versus 1.6% in non-cemented groups.
- According to NICE Health & Clinical Guidance, cemented prostheses are preferred in total hip arthroplasty (THA).

2.Physiological Models of BCIS

- Embolic model: Entry of fat, air, or cement particles into circulation.
- Histamine model: Release of histamine and local inflammatory mediators.
- Complement activation: Activation of complement pathways C3a and C5a.
- Multi-model hypothesis: Combination of the above mechanisms with simultaneous roles of several factors.

3.Preoperative and Intraoperative Precautions

Anesthesia team collaboration:

- Inform exact cement insertion timing to the anesthesia team.
- Monitor hemodynamics carefully, especially in high-risk patients.
- Optimize hemodynamics and oxygenation:
  - 100% FiO_2
  - Maintain blood pressure within ±20% of baseline
  - Availability of vasopressors

Surgical and cementing techniques:

- Modular lavage and complete drying of the femoral canal before cement insertion.
- Avoid excessive pressure in the femoral canal.
- Use the retrograde cement insertion technique to reduce air release and microembolism.

- Venting to decrease intracanal pressure and prevent embolism.
- Uniform cement mixing to improve homogeneity and reduce porosity.
- Tourniquet or similar methods to reduce local blood flow and acid release, which can cause hypothermia and local acidosis.

Sample references in APA style:

1.Donaldson, A. J., et al. (2009). Bone Cement Implantation Syndrome. British Journal of Anesthesia, 102(1), 12–20.

2.Mudalkar, N. (2011). Management of Bone Cement Implantation Syndrome. Journal of Orthopedic Research, 29(4), 567–575.

3.Brokke, K. E. (2025). Bone Cement Syndrome. ScienceDirect.

4.Yang, Y. (2024). Study of BCIS Mechanisms. Journal of Clinical Orthopedics, 13(2), 88–97.

5.So, D. (2017). Pathophysiology of BCIS. Orthopedic Reviews, 9(3), 34–42.

6.Tiago, R. (2024). Advances in BCIS Prevention. International Journal of Orthopedic Surgery, 18(1), 101–110.

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